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International Experience in the Regulation of Fusion Facilities

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INTERNATIONAL EXPERIENCE IN THE REGULATION OF FUSION FACILITIES

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INTERNATIONAL EXPERIENCE IN THE REGULATION OF FUSION FACILITIES

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FOREWORD

The IAEA recognizes the need for regulatory practices to adapt and expand in response to the evolving fusion landscape. Regulatory bodies worldwide are enhancing their expertise, leveraging the latest technical insights and refining their regulatory approaches to address the unique characteristics of fusion.

The objectives of this publication are twofold. First, it aims to comprehensively document fusion regulatory practices across Member States, including plans for future fusion power plants. It examines existing regulatory frameworks, reviews technical capabilities and outlines plans for future regulation. Second, it suggests possible areas of focus for the development and refinement of regulatory frameworks tailored to fusion facilities. Importantly, these suggestions of possible focus areas are not intended to be prescriptive but rather to serve as valuable reference points, highlighting technical and regulatory considerations that merit specific attention.

This publication provides summaries of regulatory practices and experience gained from fusion facilities worldwide. It also presents Member States' plans for future fusion regulatory approaches and discusses elements of effective fusion regulatory frameworks.

This publication is intended for regulatory bodies, policy makers and other stakeholders involved in the fusion energy sector.

The IAEA is grateful to all the contributors to this publication and to the Member States and organizations who have shared their insights and experiences. The IAEA officers responsible for this publication were M. Aoki and M. Ascic of the Division of Nuclear Installation Safety.

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1. INTRODUCTION

1.1. BACKGROUND

Several Member States are engaged in the regulation of fusion facilities. Typically, these facilities use small quantities of radioactive substances to demonstrate the feasibility of fusion technologies. Member States utilize or, when required, adapt existing regulatory frameworks for radiation protection or nuclear facilities to support the initial phases of research and development of commercial fusion facilities. Some countries have found these adaptations suitable for the short term; however, there is a need for updated or entirely new regulatory frameworks to address the increased complexity anticipated in future commercial fusion facilities. These complexities may involve managing larger quantities of radioactive material and handling extreme operational conditions, such as elevated temperatures and vacuum environments.

The International Thermonuclear Experimental Reactor (ITER) is an industrial-scale experimental facility under construction in France by the ITER members: China, the European Union, India, Japan, the Republic of Korea, the Russian Federation, and the United States of America. It is one of the few industrial-scale experimental facilities in the world. While more detailed fusion facility designs are being developed, including DEMOnstration Power Plant (DEMO) and Spherical Tokamak for Energy Production, which will demonstrate the capability to produce net electricity from fusion, there is still very limited international experience in regulating the safety of such facilities.

National regulatory bodies will continue to enhance expertise as additional technical and design information becomes available from fusion developers. This information will support the development and necessary updates for an effective regulatory framework for fusion facilities.

A related publication, Experiences for Consideration in Fusion Power Plant Design Safety and Safety Assessment, was published by the IAEA in late 2024 [1]. This publication collects practices and experiences used, or planned to be used, to ensure the safety of fusion prototype devices and fusion power plants (FPPs), along with the state of knowledge of safety approaches. The principles of design safety and safety assessment documented therein may be considered when developing the regulatory framework for fusion facilities.

1.2. OBJECTIVES

The primary objectives of this publication are to:

— Document Member States’ fusion regulation practices[1], including discussions of associated regulatory frameworks and technical capabilities, and future regulatory plans.

— Provide States with suggested areas of focus in the development and updating of regulatory frameworks for fusion facilities. The purpose of providing suggested focus areas is not to prescribe actions for States, but rather to highlight technical and regulatory topics that may warrant specific consideration to support the development of suitable and effective fusion regulatory frameworks tailored to the planned fusion facilities.

[1] Data supporting this publication were collected in 2023; subsequent changes in regulatory frameworks are not reflected in this publication. The reader is recommended to consult the respective regulatory bodies for the most current information, as policies may have been updated since the time of data collection.

1.3. SCOPE

This publication provides summaries of regulatory practices and experience in fusion facilities, highlights Member States' plans for future fusion regulatory approaches, and discusses considerations for the development of effective fusion regulatory frameworks.

This publication was developed based on the information provided by Member States on different types of fusion facility. Hybrid fission-fusion facilities are not considered in this publication. Safeguards and security considerations are also outside the scope of this publication.

1.4. STRUCTURE

This IAEA publication focuses on providing valuable insights into the regulatory landscape surrounding fusion technologies and laboratories. Section 2 presents summaries of regulatory practices in different Member States, offering a comprehensive view of their approaches. Sections 3 and 4 further explore common regulatory issues and approaches, distinguishing between prescriptive, goal oriented, and graded methods. Lastly, Section 5 addresses the development of fusion regulatory frameworks, discussing topics such as the state of knowledge, international alignment, terminology, authorization strategies, and adaptability to evolving technologies.

1.5. COMMON DEFINITIONS

For clarity and consistency, several terms used in this publication are defined as follows:

Fusion means a controlled, deliberately initiated process in which two or more atomic nuclei join together, resulting in the formation of a heavier element.

Fusion facility means fusion devices and associated equipment used for either research, experimental, or development purposes or commercial applications for the ultimate purpose of producing energy.

The terms authorization, safety, and graded approach are defined in the IAEA Nuclear Safety and Security Glossary [2].

2. SUMMARY OF NATIONAL PRACTICES

This section provides a brief description of the fusion regulatory framework, the technical capabilities of the regulatory body, stakeholder involvement in developing fusion regulatory frameworks, and plans for future fusion regulation in:

- Canada;
- China;
- France;
- Germany;
- Japan;
- Republic of Korea;
- Russian Federation;
- United Kingdom;
- United States;
- European Union.

2.1. CANADA

2.1.1. Regulatory framework

The Canadian Nuclear Safety Commission (CNSC) is the responsible authority for the regulation of fusion and fusion related activities in Canada. The mandate of the CNSC includes the protection of the health and safety of Canadians as well as the environment. This covers radiological and non-radiological risks in the conduct of regulated activities. With this mandate, anyone proposing to conduct activities that involve nuclear related activity or the possession or use of any nuclear material or substance requires a licence under the Nuclear Safety and Control Act (NSCA) [3]. The CNSC applies a mix of both performance based and prescriptive regulatory approaches in regulating the nuclear industry in Canada.

The NSCA defines "nuclear facility," and it includes "(a) a nuclear fission or fusion reactor or subcritical nuclear assembly." Therefore, a fusion reactor is regarded as a "nuclear facility." Fusion facilities are addressed in the Canadian legal framework under the Class I Nuclear Facilities Regulations [4]. That regulation states that "Class IA nuclear facility means any of the following nuclear facilities: (a) a nuclear fission or fusion reactor or subcritical nuclear assembly; and (b) a vehicle that is equipped with a nuclear reactor." The CNSC has already licensed three fusion research companies under the Class II Nuclear Facilities and Prescribed Equipment Regulations [5] (Class II licence) (typically used for lower energy accelerators) as the risk and complexity at this phase of development was considered to be very low.

2.1.2. Technical capabilities of the regulatory body

The CNSC has the in-house capabilities to conduct regulatory oversight of regulated fusion activities in Canada.

2.1.3. Early involvement of stakeholders and public in developing regulatory frameworks

As the regulatory framework is amended from time to time, CNSC broadly consults with all public, industry and government stakeholders, and Indigenous Peoples. It does this in a formal and well established manner for all requirements and guidance and will do so with regards to the future of fusion regulation in Canada.

2.1.4. Plans for future fusion regulation

Under Canadian law, future fusion facilities will be regulated either as Class IA Fusion Reactor or Class II Nuclear Facilities and Prescribed Equipment under the NSCA [3]. However, there is a realisation at the CNSC that those two options are not optimal and the CNSC is exploring the possibility of a new Class of licence to regulate future fusion facilities. Also, the CNSC is exploring the need to provide additional guidance for future fusion facilities and activities.

2.2. CHINA

2.2.1. Regulatory framework

The National Nuclear Safety Administration (NNSA) is the nuclear regulatory body of China and belongs to the Ministry of Ecology and Environment of China [6]. A fusion reactor is considered to be a nuclear facility. Some regulations are adapted to consider the differences between fusion and fission reactors.

2.2.2. Technical capabilities of the regulatory body

The technical capability of the regulatory body, NNSA, to assess and regulate fusion is mostly based on the experiences of the regulation of the safety of nuclear facilities and facilities using radioactive material. NNSA is developing technical capability specifically in the field of regulations for fusion facilities.

2.2.3. Early involvement of stakeholders and the public in developing regulatory frameworks

The Nuclear and Radiation Safety Center (NSC), the affiliated technical support organization (TSO) of the NNSA, is cooperating with other industrial and research stakeholders to develop the safety guidelines for the China Fusion Engineering Test Reactor (CFETR) and will also be involved in the safety assessment of the CFETR.

2.2.4. Plans for future fusion regulation

With assistance from the NSC, the Southwestern Institute of Physics (SWIP), the China Nuclear Power Engineering Company (CNPE) and the Institute of Plasma Physics of Chinese Academy of Sciences (ASIPP), NNSA is developing regulations for fusion facilities, which may partially use regulations for fission based facilities as a reference.

2.3. FRANCE

2.3.1. Regulatory framework

The Nuclear Safety Authority (Autorité de Sûreté Nucléaire, ASN) is the regulatory body for nuclear safety, radiation protection and transport of radioactive material based on the Transparency and Nuclear Safety Act (TSN) in France [7]. The TSN stipulates that ASN is an independent authority and provides a legislative framework for basic nuclear installations (BNIs).

Generally, there are three regulatory bodies concerning BNIs in France, each with different responsibilities:

— ASN regulates nuclear safety, radiation protection and the transport of radioactive material. ASN also regulates radioactive sources and environmental discharges within the site boundaries.

— The decentralized services of the Ministry of Labour, Regional Directorate for the Economy, Employment, Labour, and Solidarity regulate conventional health and safety at all industrial sites except nuclear power plants (NPPs) or sites where electricity is being produced. The appropriate regional directorate regulate conventional health and safety at ITER, rather than ASN, because electricity is not being produced.

— The decentralized services of the Ministry of the Ecology — the Regional Directorate for the Environment, Development, and Housing and the Departmental Directorate of Territories and the Sea — implement its policies at a regional and departmental level. A prefect acts as the state's representative within each of the 101 administrative divisions, responsible for issuing environmental permits and ensuring adherence to relevant regulatory requirements.

ITER is a BNI and nuclear safety is regulated by relevant laws, regulations and resolutions applicable to all BNIs in France.

2.3.2. Technical capability of the regulatory bodies for fusion activities

For ITER, there is one inspector working in the ASN Paris office part time and one inspector working in the ASN Marseille office part time. The Institut de radioprotection et de sûreté nucléaire (IRSN) has a central office in Paris and a local technical support office in Avignon and can lend specialists to support ASN inspections at ITER[2]. In the future, the number of people involved in the regulation of ITER is likely to increase.

IRSN has a subcontract for the physics of plasma aspects. In the past, IRSN had a contract with CERN for neutronic and structure activation calculations and with the Canadian Nuclear Safety Commission for confinement of tritium releases in the event of an accident. IRSN is conducting internal research and development programmes to develop competences in specific fields such as dust and tritium explosions and accident calculations.

2.3.3. Early involvement of stakeholders and the public in developing regulatory frameworks

The public is consulted on regulatory resolutions issued by ASN. The procedure for implementing public participation differs depending on whether the individual resolution in question is the subject of an environmental assessment. ASN is consulted, as are industrial stakeholders and the public, when the government changes legal and regulatory frameworks.

2.3.4. Plans for future fusion regulation

In France, there are no plans to modify the existing fusion regulatory framework because ITER is regulated by the existing regulations, and no new fusion projects will start in the foreseeable future. However, the regulatory body might work on guidance for future fusion facilities and activities.

2.4. GERMANY

2.4.1. Regulatory framework

At present, experimental fusion facilities are subject to regulation by the relevant regulatory body in accordance with the national Radiation Protection Act [8]. The regulatory body is the regional authority and may consult with Technical Inspection Associations and the federal government. Fusion facilities are regulated as facilities using ionizing radiation, with relevant authorizations depending on the use. The legislation for licensing nuclear facilities, the Atomic Energy Act [9], is explicitly only applicable to the use of nuclear fuels for fission (e.g. uranium, plutonium) and therefore does not apply to experimental fusion facilities or future fusion power plants.

2.4.2. Technical capabilities of the regulatory body

Local authorities regulate experimental facilities in consultation with the Technical Inspection Association, which acts as the TSO. Gesellschaft für Anlagen- und Reaktorsicherheit (GRS) is the main TSO of the Federal German regulatory body to assess fusion projects and has established a collaboration with the research centre in the field of fusion.

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[2] As of 1 January 2025, ASN and IRSN were merged to form a single Nuclear Safety and Radiation Protection Authority (ASNR).

2.4.3. Early involvement of stakeholders and the public in developing regulatory frameworks

One of the research projects funded by the federal government on the applicability of the German regulatory framework for fusion facilities is the Review of the Safety Concept for Fusion Reactor Concepts and Transferability of the Nuclear Fission Regulation to Potential Fusion Power Plants [10]. The project brought together different stakeholders (research organizations designing and operating fusion facilities and TSOs). These research activities are ongoing and there is no involvement with potential industrial vendors of fusion facilities.

2.4.4. Plans for future fusion regulation

A decision on the approach to regulating larger fusion facilities, especially those using tritium as fuel, has not yet been made as there are no plans to construct a large scale fusion facility in Germany.

2.5. JAPAN

2.5.1. Regulatory framework

The Act for Establishment of the Nuclear Regulation Authority (NRA) [11], enacted in 2012, created the NRA as Japan's new regulatory authority responsible for supervising nuclear and radiation safety. Its objectives are to protect the public and environment, as well as to safeguard Japan's national security.

Japan has no regulatory framework specifically for fusion facilities. JT-60SA, an experimental piece of equipment related to fusion, is regulated as a plasma generator based on the Act of the Regulation of Radioisotopes [12].

2.5.2. Technical capabilities of the regulatory body

NRA has not provided specific information on technical capability relating to fusion facilities.

2.5.3. Early involvement of stakeholders and the public in developing regulatory frameworks

No information on involvement of stakeholders and public was provided.

2.5.4. Plans for future fusion regulation

No information on future fusion regulation was provided by Japan.

2.6. REPUBLIC OF KOREA

2.6.1. Regulatory framework

In the Republic of Korea, the Nuclear Safety and Security Commission (NSSC), the national regulatory body for nuclear safety, is responsible for regulating fusion facilities. Fusion facilities are subject to the regulatory framework set out under the Nuclear Safety Act (NSA) [13], such as authorization, initial inspection, periodic inspection, and periodic reporting on facility use. The practical implementation of such regulation is carried out by the affiliated TSO, the Korea Institute of Nuclear Safety (KINS), as entrusted by the NSSC. The Korea Superconducting Tokamak Advanced Research (KSTAR) is operating for fusion research; it is not regulated as a specialized fusion reactor but is managed and supervised under the regulatory framework for radiation generating devices.

2.6.2. Technical capabilities of the regulatory body

The Korean regulatory authorities have sufficient experience in regulating KSTAR and several large accelerator facilities. By fully utilizing this experience, the regulatory body may consider preparing regulatory standards and procedures that reflect the unique characteristics of fusion through collaboration with the fusion communities.

2.6.3. Early involvement of stakeholders and the public in developing regulatory frameworks

No information on involvement of stakeholders and public was provided, other than the potential future collaboration with fusion communities in the preparation of new regulatory standards and procedures specific to fusion facilities and activities.

2.6.4. Plans for future fusion regulation

The detailed decision for the regulatory framework to be applied to fusion power plants, such as whether to regulate them at the same level as fission reactors, has not yet been made. Developing the regulatory framework for fusion reactors will be considered in the future.

2.7. RUSSIAN FEDERATION

2.7.1. Regulatory framework

The regulatory framework for fusion facilities containing nuclear material or radioactive substances in the Russian Federation is defined by Federal Law No. 170-FZ, On the Use of Atomic Energy [14]. If the fusion facility contains or uses nuclear material (including more than 0.2 g of tritium) it is regulated as a nuclear installation. If the fusion facility contains or uses radioactive substances, it is regulated as a radiation source. The mass and activity thresholds for radioactive material are established in regulations.

If the fusion facility uses a charged particle accelerator, it will be regulated as a generating source of ionizing radiation. Regulation (including licensing) of the use of such sources are performed in accordance with the requirements set out in Federal Law No. 99-FZ, On Licensing of Certain Types of Activities [15].

2.7.2. Technical capabilities of the regulatory body

As a nuclear regulator, Rostechnadzor has experience in nuclear and radiation safety regulation, control, and in conducting inspections at facilities containing anything from radiation sources to large NPPs. Rostechnadzor’s TSO, Scientific and Engineering Centre for Nuclear and Radiation Safety (SEC NRS), is undertaking research on developing approaches to effective safety regulation of fusion installations.

2.7.3. Early involvement of stakeholders and the public in developing regulatory frameworks

An example for the involvement of stakeholders (vendor, research organizations) in developing the regulatory framework is the Federal Project, Development of Controlled Fusion Technologies and Innovative Plasma Technologies [16], which involved State Corporation Rosatom, the national research centre Kurchatov Institute, and the Ministry of Education and Science. As part of this project, the SEC NRS and the Nuclear Safety Institute of the Russian Academy of Science were tasked with developing the regulatory framework for fusion facilities.

The process for developing new regulatory documents or amending existing ones involves publishing the draft versions on the official websites of the Russian federal authorities or the Government. This allows feedback to be gathered from the public and interested organizations before final approval.

2.7.4. Plans for future fusion regulation

The Russian Federation has proposed developing a future fusion framework with the goal of ensuring that risks are controlled to an acceptable level. The intent is for control through design, build, and operation with relevant legislation and standards. The newness of the technology as well as the transition to operational plants is acknowledged, and modelling/forecasting techniques are proposed to develop knowledge and understanding with a view to developing regulatory expectations.

The framework will be based on existing nuclear regulation infrastructure with future amendments to Federal Law No. 170-FZ [14] to define a fusion installation. The planned amendments will classify fusion facilities in terms of their intensity in addition to the mass and activity of radioactive material. Moreover, new federal rules and regulations relating to the use of nuclear energy are expected to be formulated. The regulatory body will remain the same (Rostechnadzor).

2.8. UNITED KINGDOM

2.8.1. Regulatory framework

In the United Kingdom, regulatory responsibility for fusion is divided as follows:

- Safety: Health and Safety Executive for Great Britain (HSE)[3] and Health and Safety Executive for Northern Ireland (HSENI).
- Environment: The Environment Agency in England, Scottish Environmental Protection Agency, Natural Resources Wales, and the Northern Ireland Environment Agency in Northern Ireland.

All practices involving the use of radiation, except for those established as licensed practices, are regulated by HSE and HSENI. Fusion facilities are not licensed and so HSE and HSENI regulate fusion facilities in the same way as they regulate the use of radiation for diagnostic, structural integrity, therapeutic, research, and all other practices.

Activities involving radioactive substances, such as storing or utilizing radioactive material, accumulating radioactive waste, and disposing of radioactive waste, are governed by the Schedule 23 of the Environmental Permitting (England and Wales) Regulations 2016 [17].

2.8.2. Technical capabilities of the regulatory body

The regulatory bodies have sufficient technical expertise to regulate fusion facilities and activities.

2.8.3. Early involvement of stakeholders and the public in developing regulatory frameworks

In 2021, the United Kingdom government released a green paper outlining proposed approaches for a regulatory framework for fusion energy and a separate fusion strategy [18]. The Department for Business Energy and Industrial Strategy initiated a consultation process to gather opinions regarding the recommendations on the future regulatory framework for fusion. The outcome of the consultation

[3] Regarding the transport of radioactive material, oversight is provided by the Office for Nuclear Regulation.

included recommendations that public engagement and consultation be considered by the regulatory bodies. HSE does not consult the public on its regulatory decisions with respect to fusion.

2.8.4. Plans for future fusion regulation

The outcome of the government's consultation regarding the proposed regulatory framework for fusion energy confirmed that future fusion energy facilities will fall under the existing legal framework, with regulation continuing to be provided by the Environment Agency and the HSE. Applying this regulatory approach for both safety and environmental radiation regulation to all prototype/demonstrator fusion power plants proposed for the United Kingdom will, through the lessons learned, inform any changes to the regulatory framework, if necessary. It was noted that the existing legislative definition of "nuclear installation" could provide greater clarity regarding its applicability to fusion energy facilities, thus reducing the potential for discrepancies and disruptions. The United Kingdom's government has indicated that it initiated the passage of legislation that ensures that fusion facilities will not be defined as nuclear installations.

Overall health and safety regulations and a regulatory framework are in place, but guidance will be augmented, and greater emphasis will be placed on defence in depth. Introducing specific legislation is generally not the preferred option as experience has shown that the general duties set out in the existing regulatory framework are flexible and broad enough to set minimum standards of safety.

The Environment Agency considers that the legislative framework in place is suitable for regulating both existing and forthcoming fusion technology and does not require amendments for the protection of the public and the environment. Further work is, however, needed to build capability to assess and regulate future fusion facilities (e.g. staff training, site visits, new internal and external guidance) as well as potentially developing formal pre-engagement processes to reduce risks from future fusion facilities as is already in place for fission designs. The suitability of existing guidance for regulation of future fusion power plants was considered and gaps and changes that may be required to regulatory guidance were identified.

2.9. UNITED STATES OF AMERICA

2.9.1. Regulatory framework

The regulation of fusion research and development activities in the United States of America is carried out either by 'Agreement States' using their regulatory authority assumed from the United States Nuclear Regulatory Commission (US NRC) under the National Materials Program, or by the United States Department of Energy (DOE). To evaluate Agreement States' regulatory practices, understand technologies, and develop regulatory framework options, the US NRC has established a diverse working group. The working group includes two representatives from the Organization of Agreement States, who contribute expertise based on their experience with licensing practices and regulatory oversight of fusion research and development activities. The US NRC has engaged DOE to gain insights on the operating characteristics and treatment of fusion research and development facilities under their purview.

2.9.2. Technical capabilities of the regulatory body

The US NRC is developing the necessary capabilities to review applications for future fusion facilities. This effort is an extension of capabilities needed for regulating fission reactors, fuel cycle facilities, and a wide range of uses for radioactive material. Enhancing expertise on fusion device technologies will also support the development of risk informed, performance based, and technology inclusive regulations and guidance. The US NRC is engaging with DOE, other regulatory bodies, and the country's fusion community to understand the specific technical expertise that will be necessary for regulatory safety reviews and to develop its capabilities. The US NRC will work to enhance knowledge in technical areas that are common across the array of potential fusion device designs to support future licensing reviews that implement the established regulatory framework, including:

- Tritium processing systems;
- Validation of radiation shielding analysis tools for fusion applications;
- Materials and challenges in a fusion reactor environment;
- Potential source terms and accident progression phenomena;
- Proposed design standards;
- Operating waste streams including tritium and replaceable reactor components;
- Decommissioning.

Additionally, as part of US NRC's forward focused research programme, the US NRC staff is reviewing advanced manufacturing technologies (AMTs) for fusion facility materials. The project aims to understand the technical challenges and potential regulatory gaps related to using and licensing AMT plasma facing component materials for nuclear applications.

2.9.3. Early involvement of stakeholders and the public in developing regulatory frameworks

The US NRC has collaborated with a wide variety of international partners to exchange technical and legal knowledge, best practices, and lessons learned. International partners include, among others, the United Kingdom Atomic Energy Authority; the United Kingdom Department of Business, Energy, and Industrial Strategy; and members of the ITER.

The US NRC has engaged the fusion stakeholder community on developing regulatory approach alternatives for commercial fusion energy systems. Between January 2021 and June 2022, a series of six interactive public meetings were conducted by the US NRC. To enable participative meetings and gather a broad scope of views, the US NRC solicited topics of interest from stakeholders in advance of these meetings. Stakeholder engagement has included joint workshops and pre-application meetings with private fusion companies, international collaboration through bilateral and IAEA interactions, coordination with Agreement States, and advisory committee briefings, all aimed at advancing regulatory readiness for fusion energy. From this engagement, the US NRC has gathered information on proposed fusion energy system designs and their potential hazards, as well as stakeholder perspectives on the appropriate regulatory framework.

2.9.4. Plans for future fusion regulation

The US NRC is preparing the necessary regulatory framework needed to license and regulate commercial fusion facilities in response to the Nuclear Energy Innovation and Modernization Act (NEIMA) enacted in early 2019 [19]. NEIMA requires the US NRC to:

> "(1) establish stages within the licensing process; (2) increase the use of risk-informed, performance-based licensing evaluation techniques and guidance; and (3) establish by the end of 2027 a technology-inclusive regulatory framework that encourages greater technological innovation."

The US NRC staff is leveraging lessons learned from initiatives to develop risk informed and performance based approaches for both fission reactors and materials licensees. The US NRC staff will consider the potential risks of various types of fusion facilities during the development of a regulatory framework.

The US NRC considered three options to regulate fusion facilities [20]: a "utilisation facility framework" similar to that used for operating fission reactors, a "byproduct materials framework" similar to that used for materials licensees, such as particle accelerators, or a "hybrid approach" designed to manage the varying radiological hazards linked to different fusion technologies and design concepts. Under a hybrid approach, either a bifurcated or consolidated framework would be developed. A bifurcated approach would distinguish between different fusion facilities, addressing some using a utilization facility model and others using a byproduct material model. A consolidated approach would address all commercial fusion energy systems, grading regulatory requirements based on hazards for different technologies within a single part of the US NRC regulations.

The US NRC submitted these proposed alternatives for the licensing and regulatory oversight of fusion energy systems for consideration in 2023. In the paper, titled SECY-23-001: Options for Licensing and Regulating Fusion Energy Systems [20], the prospective near-term designs for deployment in the United States of America are magnetic confinement, inertial confinement, and magneto inertial confinement technologies. For each of these technologies, the US NRC staff identified the following common characteristics that would minimise potential radiation exposure to facility personnel and the public [20]:

> "• No fissile material is present, and criticality (a self-sustaining neutron chain reaction) is not possible.
>
> • Energy and radioactive material production from fusion reactions cease without any intervention in off-normal events or accident scenarios.
>
> • Active post shutdown cooling of the fusion containing radioactive material is not necessary to prevent a loss of radiological confinement (i.e., vessel breach) ...
>
> • Radionuclides present in the fusion device, in processing or storage, or in activated materials, in any significant mobilizable amount are expected to result in low doses to workers and member of the public during credible accident scenarios (e.g., less than 1 rem effective dose equivalent to a person off site)."

In April 2023, the Commission tasked the US NRC with establishing a regulatory framework tailored to near-term fusion energy systems, building on the licensing process for byproduct material use [21].

2.10. EUROPEAN UNION

2.10.1. European Union and Euratom Treaty and legislation

The Treaty on the European Atomic Energy Community (Euratom) [22] is one of the active treaties that forms part of the European Union's primary legal framework. The Euratom Treaty is exclusively focused on the civilian applications of nuclear energy, excluding any military use. Its primary aim, within a unified nuclear energy market, is:

"• to promote research;

• to achieve security of supply for all EU countries;

• to establish a system for supervising the peaceful use of nuclear materials intended for civilian use and ensuring high common standards for health and safety."

The European Union promotes the above objectives through the:

— European Union legislation concerning radiological protection, including the Council Directives on basic safety standards [23], drinking water [24], supervision and control of shipments of radioactive waste and spent fuel [25], and Council Regulation (Euratom) on shipments of radioactive substances between Member States [26], Council Regulation on maximum permitted levels of radioactive contamination of food and feed following a nuclear accident [27], and decisions on notification and assistance in the case of nuclear accident [28, 29];

— European Union legislation on the management of radioactive waste and spent fuel, including the Council Directive on radioactive waste and spent fuel management [30];

— European Union legislation on nuclear safety, including the establishing a community framework for the nuclear safety of nuclear installations [31];

— European Union legislation on the decommissioning of nuclear installations (i.e. power plant or research reactor), including providing support for decommissioning programmes, as outlined in the Refs [32, 33];

— European Union legislation on nuclear safeguards to ensure that nuclear material is used only for peaceful purposes, including the regulation on the application of Euratom safeguards [34].

2.10.2. Interactions with European Union national regulatory bodies

The European Commission liaises with European regulatory bodies through the European Nuclear Safety Regulators Group, which is an independent expert advisory group created in 2007 [35]. It comprises senior regulators and civil servants from all European Union Member States with expertise in nuclear safety, radioactive waste, and radiation protection, along with representatives from the European Commission.

In addition, and as referred to in the Euratom Treaty Article 31 [22]:

> "The basic standards shall be worked out by the Commission after it has obtained the opinion of a group of persons appointed by the Scientific and Technical Committee from among scientific experts, and in particular public health experts, in the Member States. The Commission shall obtain the opinion of the Economic and Social Committee on these basic standards.
>
> After consulting the European Parliament, the Council shall, on a proposal from the Commission which shall forward to it the opinions obtained from these Committees, establish the basic standards;"

2.10.3. European Commission activities concerning the regulatory framework for fusion energy

In 2020–2021, the Directorate-General for Research and Innovation (DG RTD) carried out a review to examine the regulatory pathways available for fusion power plants and in 2022, the Euratom Research and Training Programme launched the HARMONISE grant towards the harmonization in licensing of future innovative fission and fusion power technologies in Europe. In 2022, EUROfusion established a working group comprising nuclear safety experts from both the fusion and fission sectors, tasked with developing recommendations for initial safety principles to guide the creation of fusion specific regulations.

The Study on the Applicability of the Regulatory Framework for Nuclear Facilities to Fusion Facilities [36] identified fusion specific safety issues and proposed safety concept for fusion facilities along with a plan to establish a fusion specific regulatory framework. The study concluded that the European secondary legislation does not address specific requirements for fusion facilities. However, due to the general character of the requirements, some would be applicable to fusion power plants. These are:

- Council Directive on basic safety standards [23]);
- Council Directive on radioactive waste and spent fuel management [30].

The following secondary legislation applies only to nuclear facilities, meaning that fusion power plants are outside of the scope since a nuclear facility is defined as "A facility (including associated buildings and equipment) in which nuclear material is produced, processed, used, handled, stored, or disposed of" [2] and the Treaty on the European Atomic Energy Community (Euratom) [34] covers fissionable materials. These include Council Directives on nuclear safety [37, 31] and the Commission Regulation on nuclear safeguards [38].

3. COMMON REGULATORY ISSUES

This is a summary of issues identified in Sections 2.1–2.9 on national practices. Related discussions can be found in Annex II. The issues include the following:

- Regulatory bodies are assessing how emerging technologies and more complex operations will fit within their frameworks for experimental fusion facilities. This will affect the authorization of the stages in a facility's lifetime, including siting, design, construction, operation, and eventual decommissioning. See Annex II, Sections II–1 to II–4.
- Either fusion is explicitly out of scope of the existing nuclear regulations because they are limited to the use of fissile materials, because the fusion facility is not clearly defined in the regulations, or because fusion is not explicitly included.
- Fusion presents a different level of hazard and associated risk than fission processes. Neither the expectations for risk control nor the approach of the regulatory body are likely to be the same for fusion as for fission.
- Regulatory guidance specific to fusion facilities does not exist.
- There is a need to explore a suitable regulatory engagement process with interested parties (vendor to regulatory body, and regulatory body to regulatory body) consistent with national arrangements, taking accounts of different levels of experience of fusion designers and other relevant national regulatory bodies. See Annex II, Sections II–5 and II–6.
- There is a need to explore a suitable public engagement and consultation process consistent with national arrangements. For example, there is a need to explore the timing and scale of public engagement to maximize the benefit to the public, developers, and regulatory bodies. Public consultations could additionally identify unintended consequences and regulatory gaps prior to implementation. See Annex II, Section II–5.

4. COMMON REGULATORY APPROACHES

Member States involved with fusion development are exploring their approach to the regulation of fusion facilities. Even though they start from different basic frameworks, some similarities have been found in the underlying concept of how Member States are initially positioned to regulate and how they will regulate in the future. This is a summary of national regulatory approaches identified in Sections 2.1–2.9 with supplemental information.

4.1. PRESCRIPTIVE AND GOAL ORIENTED APPROACHES

One way to categorize regulatory approaches for fusion facilities is by determining if Member States are following a prescriptive regulatory approach or goal oriented regulatory approach.

In a prescriptive regulatory approach, the legislation sets out detailed, specific requirements for licensees to comply with, often tailored to the particular technology used in fusion facilities or activities. This approach may specify requirements that apply to individual safety systems within the facility. Conversely, a goal oriented regulatory framework establishes overarching safety objectives, such as ensuring containment of radioactive material, leaving it to the licensee to demonstrate to the regulatory body that their design and operational methods will meet these objectives. For a comparison of the advantages and disadvantages of both approaches see Ref. [36].

In Canada, a fusion power plant would fall explicitly in the same class of facility as a nuclear fission power plant. The Canadian regulatory body is aware that this could result in imposing prescriptive or restrictive requirements. If the lower hazard potential of a fusion power plant compared to a fission power plant is credited this may require considerable efforts by applicants to justify exceptions and/or may need extensive reliance on judgements about the application of a graded approach. Even though these exceptions and/or the application of a graded approach exists, in Canada it is being investigated how to advance the regulation for fusion power plants in a more goal oriented, technology neutral way. This includes the classification of fusion facilities within the regulations [39].

The French nuclear regulation for BNIs follows a goal oriented approach focusing on performance and outcomes. Every BNI is regulated depending on the specific circumstances even if some similarities from one BNI to another occur. For ITER, a combination of prescriptive and goal oriented regulatory approaches was adopted. There is no intention to amend the regulations applicable to fusion facilities.

The existing United Kingdom regulations for fusion facilities are goal oriented and this approach will be used in the foreseeable future. This reflects the United Kingdom regulatory framework for all workplace risks where goals to be achieved are established rather than having absolute standards met.

One of the requirements of NEIMA [19] in the United States of America is for the US NRC to increase the use of “risk-informed, performance-based licensing evaluation techniques and guidance” and to “establish by the end of 2027 a technology-inclusive regulatory framework.” The use of evaluation methods and guidance that are risk informed and performance based implies that authorization will include goal oriented methods.

In practice, usually a hybrid solution between prescriptive and goal oriented approaches is used meaning that even goal oriented approaches include certain prescriptive elements to specific safety aspects that are essential for protecting workers, the public and the environment from ionizing radiation.

4.2. USE OF AND PLANS FOR GRADED APPROACHES

Several projects on fusion facilities are ongoing worldwide, covering different design options and facility sizes. Since planned fusion facilities will span from small experimental setups to large demonstration power plants, their associated radiological hazard levels will inherently be diverse.

Paragraphs 3.15 and 3.22 of IAEA Safety Standards Series No. SF-1, Safety Fundamentals [39] state:

> "3.15. Safety has to be assessed for all facilities and activities, consistent with a graded approach."
>
> "3.22. To determine whether radiation risks are as low as reasonably achievable, all such risks, whether arising from normal operations or from abnormal or accident conditions, must be assessed (using a graded approach) a priori and periodically reassessed throughout the lifetime of facilities and activities."

With regard to the regulatory framework, para. 4.62 of IAEA Safety Standards Series No. GSR Part 1 (Rev. 1), Governmental, Legal and Regulatory Framework for Safety [40] states:

> "The regulations and guides shall provide the framework for the regulatory requirements and conditions to be incorporated into individual authorizations or applications for authorization. They shall also establish the criteria to be used for assessing compliance. The regulations and guides shall be kept consistent and comprehensive and shall provide adequate coverage commensurate with the radiation risks associated with the facilities and activities, in accordance with a graded approach."

Reference [41] outlines the basic principles for applying a graded approach to the regulatory oversight of nuclear installations. It also proposes a general methodology for implementing a graded approach in the regulation of nuclear installations, supported by regulatory examples. Reference [42] emphasizes the significance of a well structured governmental regulatory framework designed to ensure protection against radiation risks. It also highlights the importance of a graded approach to match safety measures with radiation risks, offers practical guidance, and showcases Member States' examples with the aim to establish a systematic and harmonized approach consistent with IAEA safety standards.

At the core of a graded approach is the classification of a facility based on its intrinsic hazard potential, which is primarily radiological. For assessment, appropriate application of graded approach is comprehensive, encompassing all radioactive material on the site, from the tritium fuel cycle system to the facilities for waste management and storage. To classify a fusion facility in accordance with a graded approach, one way is to use the hypothetical radiological consequences determined on a postulated worst-case scenario without crediting engineered safety features (e.g. as an unmitigated risk approach). Providing guidance on the application of a graded approach within national regulatory frameworks could prove valuable for developers of fusion facilities.

In Canada using a graded approach is part of the existing regulatory framework. For example, it is applied for the regulation of research reactors. In France a graded approach is an integral part of the goal oriented regulatory framework. In the United Kingdom. a graded approach is also an integral part of the existing and planned future regulatory framework for fusion facilities. In the United States of America, graded approaches to the implementation of regulations are used in both nuclear facility and radiation protection frameworks. The United States of America is considering graded approaches for the future regulation of fusion facilities.

5. PROPOSED PRIORITY AREAS FOR THE DEVELOPMENT AND REVISION OF REGULATORY FRAMEWORKS FOR FUTURE FUSION FACILITIES

This section provides States with focus areas in the development and revision of regulatory frameworks for fusion facilities. The purpose of providing these focus areas is not to prescribe actions for Member States but rather to highlight technical and regulatory topics that may warrant specific consideration to support the development of tailored and effective fusion regulatory frameworks.

5.1. STATE OF KNOWLEDGE AND REGULATION

There are several fusion energy technologies under development internationally. The tokamak design, a type of magnetic confinement, is the most widely pursued approach. Other fusion technologies include alternative magnetic confinement designs, inertial confinement, and hybrid methods that incorporate elements of both. The fusion technology approaches are in various stages of development and engagement with national regulatory bodies has been limited. National regulatory frameworks seek to ensure an appropriate level of public health and safety, aligned with the specific nature of the facilities and activities involved. To achieve this, engagement with developers on technology attributes, safety, and operating characteristics during the development of a framework could provide regulatory bodies with valuable insights. By providing design details, developers can help regulatory bodies to anticipate the effectiveness of the proposed framework and identify unintended consequences or regulatory gaps prior to implementation. Such engagement facilitates the ability of regulatory bodies to establish appropriate regulatory frameworks for planned activities and facilities.

5.2. NATIONAL ALIGNMENT ON FUSION REGULATION

Most national regulatory bodies regulate both fusion facilities and existing fission facilities, and plan to regulate future fusion energy facilities. Historically, much of the regulatory work has consisted of fission regulation and the effect may be an internal mindset that fusion facilities can be regulated in the same way as fission facilities. There is a need for some regulatory bodies to conduct additional discussions to align on an appropriate approach to fusion regulation to provide greater clarity in regulatory requirements.

As part of the alignment of expectations and jurisdiction within an individual country, regulatory bodies may benefit from identifying whether any legislative measures are necessary. For example, in some countries, national law may dictate that future fusion facilities are authorized in a similar manner to nuclear power reactors while the regulatory body may be considering alternative approaches (e.g. use of a radiation protection framework).

Some regulatory bodies may also consider whether jurisdictional coordination is needed with regional or local governmental bodies that may be responsible for authorizing small scale research and development activities supporting fusion technology deployment. It is important that as research and development facilities scale to commercial facilities, clear jurisdiction is defined to establish the responsibilities of appropriate regulatory bodies.

It may be appropriate for the regulatory body for a fusion facility to coordinate with other regulatory bodies to establish appropriate jurisdiction over all hazards posed by the facility.

5.3. INTERNATIONAL REGULATORY HARMONIZATION

A fusion facility designer may seek an authorization for their design in more than one country. Harmonization of international regulatory frameworks or an awareness each other's regulatory frameworks would be beneficial to the designers and regulatory bodies. Harmonization of international regulatory frameworks would allow a designer to develop safety documentation that could be submitted to multiple regulatory bodies. An awareness of each other's regulatory frameworks could result in regulatory bodies providing guidance to fusion designers on what safety documentation from other regulatory frameworks may be suitable for their regulatory framework.

5.4. CONSISTENT USE OF TERMINOLOGY

Many countries have not yet established definitions for what constitutes a fusion facility in national legislation, regulations, and guidance. In some cases, the term 'fusion reactor' is used, which could incorrectly imply a fission based chain reaction. Fusion reactions do not rely on a self-sustaining neutron chain reaction; instead, they require carefully maintained conditions of heat and pressure for the reaction to take place.

Under the implementation of existing requirements in some countries, some research and development fusion facilities are treated as particle accelerators as the hazards and associated risks of the experimental fusion facilities in those countries are similar to those of a particle accelerator. Clear definitions of fusion facilities would support the establishment of a clear regulatory scope for authorization, allowing for the delineation of contents of applications tailored for specific technologies. National legislation, regulations, and guidance would also benefit from a glossary of terms for fusion that is aligned to the IAEA Safety Glossary [2].

5.5. AUTHORIZATION STRATEGIES

Existing regulatory frameworks for radiation protection and nuclear facilities employ different authorization schemes based on the complexity of technology, risk to public health and safety, environmental impact, and security implications. For example, facilities that possess and use limited quantities of radioactive material may apply for authorizations, which, if granted, would allow direct use of the material. However, nuclear facilities generally follow a more complex authorization process that could include public hearings, independent organization reviews, and issuance of multiple authorizations prior to operation (e.g. manufacturing authorizations, construction or commissioning permits, operating authorizations). Multiple authorization steps may be used for several reasons. Examples include technologies that may still be under development, pose a significant hazard to the public, or may undergo design changes during construction prior to beginning operation.

For first of a kind fusion technologies, a multi step authorization approach may promote early engagement between developers and regulatory bodies. Such engagement could help clarify regulatory expectations for developers creating new designs and to improve understanding of proposed technologies amongst regulatory bodies. For developers, a multi step authorization approach could allow for incremental regulatory approval while a design is still under development. For regulatory bodies, a multi step authorization process would also introduce hold points in the design and construction of a facility prior to operation, allowing an opportunity for regulatory bodies to confirm that regulatory requirements continue to be met prior to certain activities, including construction and installation activities, as well as a developer's receipt, use, and generation of radioactive material and activation of components.

Other authorization strategies for fusion facilities more commonly associated with commercial nuclear fission reactors, such as early approval of selected sites, approval of manufacturing facilities for mass produced devices, and design certification, may also be appropriate for regulatory bodies to consider when establishing a fusion regulatory framework.

5.6. APPLICATION OF A GRADED APPROACH TO REGULATORY FRAMEWORKS

Regardless of an individual country's selection of an appropriate regulatory framework (e.g. radiation protection, nuclear facility, or hybrid), requirements may need to take into account differences in individual fusion technologies. The regulatory authorization and oversight of existing and planned fusion research and development facilities may differ from future commercial fusion facilities when considering differences in quantities of material, types of material, physical reactions, and use of technologies at various facilities. Appropriate scaling of requirements may take into consideration, but is not limited to, the following:

- Hazards posed to facility workers;
- Impacts to members of the public and environment;
- Siting considerations;
- Differences in operational aspects (e.g. need for operating personnel);
- Different programmatic aspects;
- Emergency planning;
- Physical security;
- Cyber security;
- Waste management;
- Decommissioning.

Appropriate scaling of regulatory requirements may also take into consideration uncertainty associated with potential operational and accident conditions that could pose a risk to workers and the public. Such consideration of uncertainty may include regulatory requirements that impose conservative analytical margins or additional physical attributes (e.g. shielding, confinement, containment) until the state of knowledge of fusion technologies improves to justify the removal of such mitigating and preventative measures.

5.7. DEVELOPMENT OF IMPLEMENTING GUIDANCE

As part of the establishment of a regulatory framework, the development of implementing guidance may be undertaken to aid developers in the preparation of applications and regulatory bodies in the review of submitted applications. Existing guidance documents for radiation protection and nuclear facilities were, generally speaking, not created for the regulation of fusion facilities. To the extent practicable, guidance for fusion facilities would benefit from being performance based to establish clear safety goals while allowing maximum flexibility to accommodate a wide array of fusion technologies. However, guidance documents may establish acceptable means of hazard analysis, content of applications, and necessary technical detail to support predictability in regulatory reviews. As part of updating existing guidance documents or creating new guidance documents, particular attention may be focused on, but not limited to, the following technical areas that may present unique considerations for commercial fusion facilities:

- Safety and hazard analyses;
- Tritium management;
- Physical security;
- Cyber security;
- Waste management;

— Emergency preparedness;
— Radiation protection;
— Decommissioning;
— Siting.

There is a lack of fusion specific international codes and standards for designers to reference as relevant good practices during the safety demonstration of their design. Fusion specific international codes and standards would be a sound starting point for demonstrating best practices. There are codes and standards for nuclear facilities, although their use could be disproportionate to the risk posed by the operation of fusion facilities.

5.8. EVOLVING TECHNOLOGIES AND REGULATORY BODY AGILITY

The continued development and deployment of fusion facilities will depend, in part, on regulatory bodies that are prepared to independently analyse and evaluate the safety implications of the new technologies. Emerging technologies from other industries promise to support the fusion industry by providing safety and economic benefits that underpin both existing and planned facilities. For example, the fusion industry is benefitting from the adoption of new technologies used in other industries, such as artificial intelligence and autonomous operation.

The ability of regulatory bodies to adapt and prepare for inclusion of emerging technologies in fusion facilities will be essential to proactively address potential regulatory challenges. Regulatory challenges that can be identified and planned for and are considered ‘known unknowns.’ However, there may also be unanticipated challenges that may only be identified during or after the initial authorization of a fusion facility. These challenges are referred to as ‘unknown unknowns.’

Regulatory bodies can prepare for ‘known unknowns’ and limit the number of ‘unknown unknowns’ by continuing regular engagement among fusion developers, industry groups, other national and international regulatory bodies, and research organizations. Such engagement will promote the early identification of new challenges such that they can be addressed promptly and mitigate potential delays in shaping regulatory frameworks or in the authorization process.

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ANNEX I.
SUPPLEMENTARY EXAMPLES AND INFORMATION FOR THE REGULATION OF FUSION FACILITIES

The practical examples provided in this annex capture additional information on the status of and planning for the development of fusion regulatory frameworks as of the time of submission. The information provided in this annex includes a non-exhaustive selection of existing and planned fusion facilities for illustrative purposes and to contextualize the regulatory approaches discussed. These examples are part of the global fusion research and experimental activities; more detailed records on this topic are maintained separately by the IAEA to track technological progress and facility parameters[4].

I–1. CANADA

I–1.1. Existing and planned fusion facilities

General Fusion (GF) holds a licence that authorizes testing with deuterium only [I–1]. GF is working on the development of a magnetised target fusion system, which applies both magnetic and inertial confinement. Plasma is injected into a magnetised conducting vortex of molten lithium-lead. An array of pistons around the molten lithium-lead drives a pressure wave into the centre, compressing the plasma to achieve fusion conditions. This cycle is repeated, with the resulting heat absorbed by the liquid metal and converted into electricity through a steam turbine. The inventory of tritium in the device is $<10^{15}$ Bq (~3g) and maximum energy per particle is <50 MeV.

I–1.1.1. Planned fusion facilities and fusion power plant

GF is planning a Fusion Demonstration Power Plant project, for which development and testing licences have been issued. The first is for a physics experiment investigating fusion in the solid state. This is an extremely low risk facility creating perhaps one neutron every four seconds. The second is for investigating sheared axial flow stabilised Z pinch devices, with projections for 10^{11} neutrons per shot.

I–1.2. Regulatory framework and planned regulatory approach

I–1.2.1. Regulatory framework

The mandate of the CNSC includes the protection of Canadians, as well as the environment. This covers radiological and non-radiological risks in the conduct of regulated activities. With this mandate, anyone proposing to conduct activities that involve the generation of nuclear energy would require a licence under the NSCA [I–2].

As noted in Section 2.1, fusion facilities are addressed in the Canadian legal framework under the Class I Nuclear Facilities Regulations [I–3]. CNSC has licensed three fusion research companies under the Class II Nuclear Facilities and Prescribed Equipment Regulations [I–4] (Class II licence). Significant differences also exist with regard to security requirements and the amount of financial

[4] See World Survey of Fusion Devices 2022: https://www.iaea.org/publications/15253/world-survey-of-fusion-devices-2022. For the most recent annual progress and facility updates, see the IAEA World Fusion Outlook 2025: https://doi.org/10.61092/iaea.agh6-zpih. In addition, see IAEA Fusion Portal, including Fusion Facility Database, for regular updates and future data: https://nucleus.iaea.org/sites/fusion-portal/SitePages/Home.aspx.

liability required. In addition to GF, two other companies hold a Class II development and testing licence.

As the regulatory framework is periodically revised or updated, CNSC broadly consults with all public, industry and government stakeholders and it has done so regarding the regulation of fusion facilities in Canada.

I–1.2.2. Planned regulatory approach for future fusion facilities

Future fusion facilities will be regulated either as Class IA Fusion Reactor or Class II Nuclear Facilities and Prescribed Equipment, or potentially as a hybrid of the two. A possible criterion differentiating fusion activities licensed under the Class II regulations compared to the Class I Nuclear Facilities Regulations (e.g. Class IA fusion reactor licence) could be net energy generation and/or the potential effects to the public under postulated accident conditions. Consideration will also need to be given to the nature of components that could become activated over time.

Unlike a fission reactor, in a pure fusion reactor concept, there is no possibility of out-of-core criticality. This would suggest that although fundamental safety objectives would remain the same, the conduct of activities associated with pure fusion facilities would allow additional flexibility in applying these safety objectives. However, there are countries who are developing hybrid fusion/fission concepts, so CNSC will take this into account in its regulatory strategy. This would suggest not separating fission and fusion regulatory practices but rather keeping the existing 'technology neutral' approach to the extent practicable with additional guidance for specialized applications.

I–1.2.3. Preparatory study for regulation over fusion facilities

The CNSC, in collaboration with an independent nuclear consultancy, has carried out a study to assess fusion and its implications for the CNSC. The CNSC study produced six key deliverables, including hypothetical scenarios outlining potential fusion facility hazards, international regulatory comparisons, and a baseline review of the CNSC's existing regulatory framework. It also included an initial assessment of regulatory preparedness, potential licensing and implementation challenges, and the CNSC's capability to review fusion applications. The final report provided an overall evaluation of readiness, highlighting specific regulatory and implementation issues related to emerging fusion technologies.

As the CNSC continues to refine its approach to regulating fusion technology, the report provides guidance to build on initial findings. Key suggestions include recognizing that fusion differs significantly from fission, both in technology and risk profile; the importance of understanding fusion thoroughly; and avoiding a fission based regulatory mindset. It suggests setting high level objectives, using a graded approach, and progressively developing fusion specific guidance, from simple to complex. Additionally, it highlights the potential value of international regulatory harmonization.

The report identifies several key focus areas for regulating fusion technology. These include radiological concerns associated with neutron radiation, tritium, and activated materials; chemical hazards involving substances such as beryllium and liquid lithium; thermal risks from plasma and cryogenics; physical hazards including rotating machinery and magnetic forces; electrical dangers from high voltage and current; and other considerations such as radio frequency fields, flammable materials and hydrogen use.

There is a question of whether to distinguish between facilities that use tritium and those that do not: to date no Canadian fusion facility has used tritium. The CNSC will also need to address the fact that it already licenses neutron generators that use tritium; in Canada, these are treated as accelerators. If a graded approach is applied, are facilities that do not use tritium to be considered as Class IA?

It is possible that the focus needs to be on a more generalised approach that excludes any fusion facilities, accelerators and laboratories (e.g. neutron generators) from being Class IA or magnetic confinement tokamak types. The preference of the CNSC is a more general approach, to cover the broadest scope of fusion facilities.

CNSC established a Fusion Coordination Team to create an information source within the CNSC on matters related to fusion technology, safety, regulation, and also to assess regulatory readiness for future fusion projects in Canada. The mandate of the Fusion Coordination Team is to:

- Share and compile information on existing and potential technology variations of fusion installations.
- Create an electronic sharable library of materials related to fusion regulation.
- Analyse the potential facilities to be built in Canada in terms of types and size and devise a suitable classification from the nuclear regulatory point of view.
- Develop a fusion technology licensing and compliance oversight strategy proposal for implementation at the CNSC.
- Develop an implementation plan once the strategy is approved.
- Develop an accessible, transparent location where all CNSC staff can learn about the fusion related international work underway at the CNSC (i.e. working groups, conferences) and will have access to the associated documentation/presentations/reports.
- Maintain an up-to-date internal and external information summary report/webpage on the status of the fusion technology and CNSC readiness for regulation.
- Engage in communication with major international entities active in the development and regulation of fusion facilities.
- Suggest participation in bilateral and international activities and events with peer organizations.
- Share information gathered from various international and national interactions related to fusion.
- Prepare a presentation to senior management on the status of the fusion technology and the CNSC readiness for fusion regulation.

The Fusion Coordination Team has been engaged with the IAEA on the following:

- Consultancy: A review of regulatory approaches for fusion facilities and to identify common gaps and common approaches.
- Draft publication: A compilation and review of practices in safety assessment and safety design for FPPs.

I–1.3. Technical capability in the regulatory body

The CNSC has the capabilities to conduct regulatory oversight of regulated fusion activities in Canada, with a focus on prototype facilities. Regulation of these facilities can draw from existing expertise from other already regulated activities such as nuclear installations, accelerators, and processing facilities.

With respect to new proposals, the CNSC is engaged with fusion technology developers, applying the pre-licensing process described in Ref. [I–5]. This process solicits proposal information to develop a suitable, risk informed assessment strategy. The outcome will guide applicants in preparing licence applications for their specific projects. The process is intended to be iterative, involving multiple exchanges between the CNSC and the applicant before finalising the strategy. Conduct of this process increases CNSC awareness of specific technologies. Here the CNSC can determine where the required expertise resides in the CNSC or where potential technical needs could be addressed through training or external technical support.

Overall, the CNSC possesses general and specialist expertise to support regulatory functions across all activities subject to regulation under the NSCA. Crucial to this capability is not only technical proficiency but also a clear understanding of the underlying objectives of regulatory requirements and guidance, and how these can be effectively applied to various technologies.

I–1.4. Regulations and guides including those unique to fusion facilities

Fusion facilities will be regulated either as Class IA Fusion Reactor or Class II Nuclear Facilities and Prescribed Equipment under NSCA [I–2]. Reference [I–6], describes CNSC's role, environmental assessment, and licensing process and states:

> "The following regulations made under the NSCA list the information applicants must submit to the CNSC as part of their licence applications:
>
> - General Nuclear Safety and Control Regulations
> - Radiation Protection Regulations
> - Class I Nuclear Facilities Regulations
> - Class II Nuclear Facilities and Prescribed Equipment Regulations
> - Uranium Mines and Mills Regulations
> - Nuclear Substances and Radiation Devices Regulations
> - Packaging and Transport of Nuclear Substances Regulations
> - Nuclear Security Regulations
> - Nuclear Non-proliferation Import and Export Control Regulations
>
> "A licence application must be accompanied by payment of the application fee as set by the *Canadian Nuclear Safety Commission Cost Recovery Fees Regulations*.
>
> "Applicants must also be aware of, and comply with, other federal, provincial or territorial, and municipal legislation that may also apply to their projects.
>
> "The CNSC issues regulatory documents that set out requirements and guidance that many applicants may use in preparing their licence applications. Regulatory documents form the basis for the assessment of licence applications. These documents are developed through a transparent consultative process with stakeholders, which include the general public, licensees, other government agencies, and non-governmental organizations."

A Class II nuclear facility comprises the designated equipment, the structure (or relevant parts of it) that houses this equipment, and any supporting systems required for its operation. The CNSC mandates certification of this Class II equipment to confirm that its procurement, installation and operation comply with Ref. [I–4]. These certification requirements are independent from those set by other authorities. Reference [I–7], describes the certification process, completion of the application form and cost recovery.

From a forward looking perspective, a possible criterion to differentiate fusion activities licensed under class II regulations compared to Class I regulations could be net energy generation and/or handling and use of hazardous materials and potential effects to the public under postulated accident conditions. Consideration will also need to be given to the nature of components that could become activated over time.

I–1.5. Licensing process

I–1.5.1. Licensing process based on the relevant regulation

Class IA nuclear facilities follow a stepwise licensing process that includes obtaining a licence to prepare the site, a licence to construct, a licence to operate, and finally, a licence to decommission. Class II nuclear facilities are also subject to a stepwise licensing process, which begins with a licence to construct[5], followed by a licence to operate and a licence to decommission. The CNSC regulatory documents in Refs [I–8 to I–10] are applicable.

The CNSC's licensing processes are proportional to the risk of the activities proposed taking into account the qualifications of the licensee, the safety and control measures established by the licensing, and the uncertainties associated with novel approaches and technologies. The CNSC works with a potential applicant as part of pre-licensing engagement to select the licensing process that is most appropriate for the proposed activities. The licensing process then evaluates the proposed safety and control measures and develops a compliance strategy which is informed by the safety case and the licensee's ongoing compliance history. The CNSC has corporate wide processes for graded enforcement of compliance with the licensing basis.

I–1.5.2. Pre-licensing Vendor Design Review

The CNSC has a pre-licensing Vendor Design Review (VDR) process [I–11], as follows:

(a) Aim:

A VDR is an optional process that does not involve any formal decision by the Commission under the NSCA that enables a vendor to anticipate and address project risks prior to licensing. A VDR assesses whether the vendor's design intent reflects an understanding of Canadian regulatory expectations and terminology. In this process, the vendor needs to be prepared to demonstrate how systematic and quality assured design activities are being conducted, with a well documented rationale for design and safety analysis decisions.

(b) Processes:

(i) Review process:

This review neither certifies a reactor design nor results in the issuance of a licence under the NSCA, and it is not a mandatory step in the licensing process for a new NPP. The outcomes of any design review are non-binding and do not affect the decisions made by the Commission.

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[5] Not strictly true for development and testing licences. Because the equipment can evolve, the shielding and safety systems needed (usually considered as part of construction) can also evolve so there is no way to separate the construction and operation phases. However, a "commissioning" approach is always required for any development and testing of radiation emitting devices.

(ii) Regulatory outcome:

The purpose of a review is to assess, at a high level, whether a proposed new type of facility aligns with Canadian nuclear regulatory requirements, expectations, and applicable codes and standards. These reviews also help to identify any fundamental obstacles to licensing the design in Canada and ensure that there is a clear path to resolving any issues that may arise.

(iii) Public engagement and consultation:

As a pre-licensing VDR does not result in a formal regulatory decision, CNSC staff maintain a balance between protecting commercially sensitive information provided by the vendor and ensuring sufficient access to conduct an effective review. At the same time, efforts are made to communicate the CNSC's activities to the public as transparently as possible. Accordingly, the CNSC publishes an executive summary presenting the overall conclusions and key findings of the review for public information. However, the disclosure of detailed discussions and review results remains at the vendor's discretion.

(c) Regulatory capacity and capability (define size and skills of regulatory team):

Multidisciplinary team across different specializations. The size and composition of the regulatory team is determined upon receipt of a preliminary description of the proposed activities.

(d) Duty holder submission/documentation (including scope):

Regulatory guidance on the process and information to consider in vendor design review submissions is included in Ref. [I–11].

(e) Regulatory cost recovery:

A service agreement involving the CNSC and a reactor vendor establishes the terms and conditions including cost recovery of efforts expended by CNSC staff.

The challenge with conducting a VDR on a fusion facility is that there would be a significant upfront effort for both the CNSC and a fusion reactor vendor to determine how the existing regulatory documents related to design and safety analysis could be applied to their proposed fusion facility. The fusion reactor design would also have to be sufficiently developed to facilitate the review.

I–1.5.3. Early engagement with the proponent

(a) Aim:

Assess the proposed activity in terms of its uniqueness, complexity, and potential risks, and provide the proponent with a summary of the relevant regulations, regulatory guidance, and the documentation required to support licensing.

(b) Regulations:

The applicable regulations vary depending on the activities proposed. The objective of the CNSC pre-licensing engagement process is to determine which existing regulations apply to a specific fusion facility.

(c) Processes:

(i) Engagement:

This optional process undertaken prior to submitting a licence application and after a proponent has described the conceptual project hazards in a preliminary description.

(ii) Review Process:

A proponent submits a preliminary description (conceptual level) of the proposed activities and potential hazards is submitted. The submission is assessed by CNSC staff with conclusions and recommendations documented in a draft report. The draft report is reviewed, and an assessment strategy is decided on by CNSC management and technical experts.

(iii) Regulatory outcome:

No decision is made by the Commission under the NSCA. The CNSC lead licensing director formally responds with a supplemental guidance letter which provides a summary of relevant regulations, guidance documents, and the required information to support a licence application A regulatory decision will be made after this process has been completed.

(d) Public engagement and consultation: Under discussion at the CNSC.

(e) Regulatory capacity and capability (define size and skills of regulatory team):

Multidisciplinary team across different specializations. The size and composition of the regulatory team is determined upon receipt of a preliminary description of the proposed activities.

(f) Regulatory guidance:

Regulatory guidance on the process and information to consider in a preliminary description of the proposed activities is included Ref. [I–12].

(g) Duty holder submission/documentation (including scope):

The initial outline or summary of activities and associated hazards submitted by the proponent needs to cover the full lifetime of the potential project in enough detail to enable CNSC staff to begin a technical assessment. This assessment will help identify regulatory considerations and inform a proposed licensing strategy. The description is expected to clearly outline the nature of the activities and the potential hazards they may pose to workers, the public, and the environment.

(h) Regulatory cost recovery: Under discussion at the CNSC.

I–1.5.4. Regulatory approach for planned fusion facilities

At this time there is no specific licensing process for fusion facilities. GF has announced plans to pursue a Fusion Demonstration Plant project. Given the novelty of the project, the CNSC has reached out to GF suggesting pre-licensing engagement to recommend the appropriate set of regulations and regulatory framework documents to consider in the information to be submitted in support of licensing. At time of initial licensing, the licensee's proposed safety case was evaluated and verified by staff. The CNSC analysis included independent Monte Carlo simulations. The licensee's protocol requires staff to maintain a 100 m safety radius that would result in dose to staff equivalent to background levels. In the worst case, the tritium production and activation would be below the unconditional clearance levels as stipulated in Ref. [I–13].

I–1.6. Regulatory inspection of fusion facilities

The CNSC conducts regular inspections and assessments to ensure that licensees are meeting legal and regulatory requirements, along with the specific conditions of their licences. This oversight helps confirm that licensees are operating safely and in full compliance. Compliance is verified through on-site inspections, reviews of operational activities, and examination of licensee documentation. Licensees also report routine performance data and any unusual incidents. Furthermore, the CNSC investigates unplanned events or accidents involving nuclear materials or substances and collects samples for analysis in a Canadian laboratory.

I–1.7. Stakeholder involvement

In Canada the main difference between a Class I facility and a Class II facility is that the former involves a public hearing process (generally issued by the Commission), whereas the latter does not (generally issued by a designated officer). This has an impact on the timeline for licensing. In response to any amendments to the regulatory framework, the CNSC formally consults with all relevant stakeholders.

I–2. CHINA

I–2.1. Existing and planned fusion facilities

I–2.1.1. Existing fusion facilities

There are the following existing fusion facilities:

- Sino-United Spherical Tokamak: Operated by Tsinghua University since 2002.
- Joint Texas Experimental Tokamak: Operated by Huazhong University of Science and Technology since 2007.
- HL-2M: Operated by China National Nuclear Corporation, Southwestern Institute of Physics, since 2020. HL-2M aims to generate plasmas hotter than 200 million °C and is the successor of HL-2A.
- Experimental Advanced Superconducting Tokamak: Operated by Hefei Institutes of Physical Science, Chinese Academy of Sciences, since 2006. This project aims to demonstrate relatively long (>100s pulse duration) high performance (H-mode) plasmas and is the successor of HT-6M.

I–2.1.2. Planned fusion facilities

CFETR is planned to demonstrate steady state operation with tritium self-sufficiency. It is planned for 200 MW fusion power in Phase I and 1 GW in Phase II.

I–2.2. Regulatory framework and planned regulatory approach

China has established and is improving the system of laws and regulations for nuclear facilities covering radiological safety, conventional safety, security, safeguards, and the environment. The main laws related to this field are:

- Nuclear Safety Law of the People’s Republic of China [I–14].
- Law of the People’s Republic of China on the Prevention and Control of Radioactive Pollution [I–15].

— Law of the People's Republic of China on Work safety [I–16].

— Law of the People's Republic of China on Prevention and Control of Occupational Diseases [I–17].

As a civil nuclear facility, the fusion reactor will follow these general laws and regulations for a nuclear facility.

The NNSA, which is nuclear regulatory body of China and belongs to the Ministry of Ecology and Environment of China, preliminarily considers fusion reactors as nuclear facilities. The general laws and regulations for nuclear facilities are applied to fusion reactors although some regulations cannot be simply adapted to fusion reactors because of the obvious differences of safety characteristics between fusion and fission.

NNSA is developing fusion regulations which may partially take fission regulations as a reference, assisted by the NSC, SWIP, China Nuclear Power Engineering Company (CNPE), and ASIPP. A collaborative project to study the nuclear safety of the CFETR as it pertains to China's regulatory framework is being implemented by SWIP, CNPE and ASIPP. The findings from this project are expected to provide valuable input for the creation of fusion regulations in China.

Now, SWIP is operating the HL-2A tokamak experimental facility and is constructing the HL-2M tokamak experimental facility with its collaborators. HL-2A and HL-2M are radiological facilities defined by Regulations on Safety and Protection of Radioisotope and Radiation Installations (HAF 800) [I–18]. According to the classification rules of radioactive installations made by the Ministry of Ecology and Environment of China, HL-2A belongs to Category III class with low hazards.

The Regulations on Safety and Protection of Radioisotope and Radiation Installations (HAF 800) [I–18] mostly focus on the hazards of electromagnetic and X ray radiation, which does not cover the hazards related to fusion, such as nuclear safety, and the inventory and release of radioactive substances.

I–2.3. Technical capability in the regulatory body

The technical capability of the regulatory body, NNSA, to assess and regulate fusion is mostly based on the experiences of the regulation of fission safety and radioactive material. The NSC, as the affiliated TSO of the NNSA, has been involved in the fusion safety field, and is cooperating with SWIP, CNPE and ASIPP to develop the safety guideline for CFETR, and will also be involved for the safety assessment of CFETR.

I–2.4. Regulations and guides including those unique to fusion facilities

Since 1982, China has progressively developed its nuclear safety regulatory system, drawing on IAEA Safety Standards and closely examining the nuclear safety laws and regulations of other countries with nuclear power programmes. In addition to Refs [I–14] to [I–18], the key laws and regulations include the following:

— Regulation on Safety Supervision and Administration for Civil Nuclear Facilities (HAF 001) [I–19].

— Regulation on Safety Supervision for Nuclear Facilities (HAF 001/02) [I–20].

— Regulation on Safety for Civil Nuclear Fuel Cycle Facilities (HAF 301) [I–21].

— Regulation on Safety Administration for Radwaste (HAF 400) [I–22].

- Regulation on Nuclear Material Control (HAF 501) [I–23].
- Regulation on Supervision and Administration of Civil Nuclear Safety Equipment (HAF 600) [I–24].
- Regulation on Safety Supervision and Administration for Civil Nuclear Pressure Equipment (HAF 601/01) [I–25].
- Regulation on Transportation Safety Administration of Radioactive Substance (HAF 700) [I–26].
- Regulation on Environment Protection by Electromagnetic Radiation (HAF 901) [I–27].

I–2.5. Licensing process

I–2.5.1. Licensing process in place

The fusion reactor will follow the general laws and regulations for nuclear facilities. The generic outline of the process is provided below:

(a) Permission to start the project design (project feasibility study and site confirmation):
   (i) Scope: conceptual design, site survey and recommendation, project proposal report.
   (ii) Applicants' submission: project proposal report, project conceptual design report.
(b) Permission for the construction of the nuclear facility:
   (i) Scope: site evaluation, safety analysis (feasibility study stage and design phase), environment impact assessment (feasibility study stage and design phase), project feasibility study.
   (ii) Applicants' submission: safety analysis report (feasibility study stage and design phase), environment impact assessment report (feasibility study stage and design phase), project feasibility study report.
(c) Permission of the first fuelling:
   (i) Scope: application for civil nuclear material usage, safety analysis (final stage), quality assurance for operation.
   (ii) Applicants' submission: application report for civil nuclear material usage, final safety analysis report, quality plan for commissioning and operation.
(d) Approval for the decommissioning and delicensing:
   (i) Scope: decommissioning study, environment impact assessment report for decommissioning.
   (ii) Applicant's submission: decommissioning report, environment impact assessment report for decommissioning.

I–2.5.2. Regulatory interaction with duty holders

The NSC, the affiliated TSO of the NNSA has been involved in the fusion field and is cooperating with SWIP, CNPE and ASIPP to develop the safety guideline for the CFETR and also will be involved for the safety assessment of CFETR.

Based on an assessment, due to the difference between fission reactors and fusion reactors, the regulations adapted to fission reactors cannot be fully applied for fusion reactors except for some general regulations for a nuclear facility. The development of regulations for fusion reactors will be based on the fusion theory principle, its safety characteristics and hazards, and the experience of ITER.

I–2.6. Regulatory inspection of fusion facilities

Routine inspection, regular inspection, and special inspection are conducted covering:

- Operating procedures;
- Quality assurance procedures;
- Operational measurements and reports;
- Maintenance results and reports;
- Periodic reports;
- Notification of significant non-routine activities;
- Report of abnormal events during operation;
- Emergency reports of nuclear accident.

I–3. FRANCE

I–3.1. Existing and planned fusion facilities

The W Environment in Steady-State Tokamak has been in operation under the French Alternative Energies and Atomic Energy Commission since 1988, with facility upgrades carried out in 2006 and again in 2018. This facility uses superconducting magnets; tungsten coated tiles as the first wall material and has long pulse durations.

ITER is under construction as an international project. The authorized operating domain (tritium ≤ 4 kg in whole facility; ≤ 1 kg in vacuum vessel; dust in vacuum vessel ≤ 1000 kg tungsten and beryllium). ITER aims to achieve a fusion energy gain factor of at least 10, measured using heating input and thermal output. A construction licence was issued in November 2012.

I–3.2. Regulatory framework and planned regulatory approach

I–3.2.1. Regulatory framework

ASN serves as the regulatory authority for nuclear safety, radiation protection, and the transport of radioactive material, operating under the framework of Ref. [I–28]. The TSN stipulates ASN as an independent authority and provides a legislative framework for BNI. Under TSN, ASN can authorize the creation of a BNI, decommissioning and other activities.

The TSN Act also defines the facilities that can be considered BNIs, as follows:

- Nuclear reactors;
- Facilities preparing, enriching, fabricating, processing, or storing nuclear fuel or treating, storing or disposing of radioactive wastes;
- Facilities containing radioactive or fissile substances meeting specific criteria;
- Particle accelerators meeting specific criteria.

An inventory of 1×10^{16} Bq of tritium (around 27 g) is enough to classify a facility as a BNI, specified by Ref. [I–29].

I–3.2.2. Regulatory experiences from ITER

ITER is classified as a BNI based on the amount of tritium. ITER is a different reactor compared to other BNIs in France and its specifications are taken into account in reviewing the licence applications. The following are considered:

— Application of the principle of proportionality as set out in Ref. [I–30];
— Consideration of long duration of plasma pulses (7 minutes) compared to other radiation generating facilities (e.g. medical application);
— Respect of specific limits as stipulated in safety regulations (e.g. dose limits in the event of an accident);
— Preference of a goal setting approach focusing on performance and outcomes. For ITER, a balanced approach was adopted, combining both prescriptive and goal setting regulatory approaches.

The tokamak and tritium buildings at ITER are well advanced. The ITER tokamak is expected to apply a staged approach to operation, initially creating a plasma (without fusion and hence without significant radiological hazards) and may also be operated in modes which do not require tritium as the fuel, for example using deuterium-deuterium reactions. This means that there would be limited radiological hazard in the first phase, until the tritium is added later.

ASN's Paris and Marseille offices are both involved in controlling and regulating the ITER facility, with the Paris office reviewing safety files and the Marseille office carrying out field inspection of the ITER site in Cadarache.

I–3.2.3. Other regulatory bodies

Generally, there are three regulatory bodies in France, each with different responsibilities, as listed in Section 2.3.1.

I–3.3. Technical capability in the regulatory body

IRSN provides technical support for ASN. For example, IRSN undertakes detailed technical assessments of safety files for ASN and provides detailed technical reports including recommendations. IRSN often engages directly with the licensee during a safety assessment. Once IRSN has issued its technical report, a dialogue takes place between ASN and IRSN. ASN then uses the information provided by IRSN to take a final position. ASN and IRSN documents are public.

IRSN also provide technical support to ASN during inspections. For example, specialists from IRSN may also go on site with ASN (e.g. a specialist Civil Engineer from IRSN would go on-site with ASN during the construction phase).

For ITER, there is one inspector working in the ASN Paris office part time and one inspector working in the ASN Marseille office part time – see Section 2.3.2. In addition, IRSN has a subcontract for the physics of plasma aspects – see Section 2.3.2.

I–3.4. Regulations and guides including those unique to fusion facilities

The creation of an BNI is mainly covered by the Article 29, and is also referred to in Articles 1, 3, 22, 32, 33, 45, 48 and 51, of Ref. [I–28]. The technical requirements governing BNIs include orders issued by ministers and jointly by multiple ministries, regulatory decisions made by ASN, fundamental safety rules along with ASN's guidance, and industry specific codes and standards established within the French nuclear sector.

While France lacks dedicated fusion energy regulations, facilities such as ITER are regulated as BNIs by the ASN through specific, case by case decisions within the established nuclear safety framework. Moreover, there is no information available to indicate that the regulation applicable to fusion will change in the future.

I–3.5. Licensing process

I–3.5.1. Authorization procedures for BNIs

The licensing process for BNIs involves several distinct stages and procedures, as follows:

(a) Safety options review: Before formally starting the authorization process, an operating organization planning to establish a BNI may request ASN's opinion on some or all the proposed safety options for the facility. ASN may recommend additional studies or justifications, which can support the future authorization application. These options are later included in the preliminary safety analysis report as part of the application. Although this is not a substitute for regulatory review, it helps streamline the later stages.

(b) The creation authorization: ASN assesses applications for creating or decommissioning a BNI and submits a draft decree to the Government. This decree outlines the safety, environmental, and risk prevention requirements associated with the facility.

(c) The public inquiry and environmental assessment.

(d) Creation of a local information committee to facilitate communication and provide local stakeholders with information about the BNI.

(e) Engagement with other European Union Member States: Other Member States are consulted where relevant, in line with international obligations.

(f) Technical Consultation: The preliminary safety analysis report, submitted with the authorization application, is submitted to ASN and referred to one of its Advisory Committees for expert input.

(g) The creation authorization decree: The Minister in charge of nuclear safety issues the decree after receiving input from the ASN.

(h) ASN requirements for creation authorization decree implementation: ASN defines specific requirements relating to the BNI's design, construction, and operation. It also sets conditions on water intake and environmental discharges. Any proposed changes that could significantly increase discharges or water use are to be made publicly available. These requirements are subject to ministerial approval. Reference [I–31] outlines the public consultation procedure for such changes.

(i) Commissioning authorization: Commissioning marks the first use of nuclear material in the BNI. To obtain authorization, the licensee submits updated safety documents, operating rules, a waste management study, emergency response plans, and a decommissioning strategy. ASN issues the final commissioning approval.

(j) Modification of an BNI: Any major modification to a BNI undergoes a similar process to that required for initial creation authorization.

(k) Other facilities within BNI perimeter: Equipment within a BNI boundary that falls under different administrative regimes remains governed by those regimes. However, ASN retains authority to impose and oversee individual measures.

(l) Decommissioning decree: Decommissioning is formalised through a decree, issued following ASN consultation. Once decommissioning is complete, the facility may be delicensed. In France, each BNI is assessed individually, with no standardised design framework, although some features may be shared across installations.

I–3.5.2. Regulatory experiences over ITER

In 2008, an application was submitted to ASN for ITER to become a BNI, which was rejected. After having reviewed and resubmitted its application, ASN considered the file acceptable. At the end of 2010, ASN launched a technical assessment which was undertaken by IRSN. Following this assessment, a construction licence was issued for ITER via Ref. [I–32].

There are a number of regulatory hold points, which have to be approved by ASN for onward construction of the plant. IRSN considered that the main risks of a fusion facility are tritium discharges to the environment, potential hydrogen explosion and evacuation of thermal power. Other significant risks of a fusion facility for workers include the fast neutron flux from operation (which could result in a potentially fatal operator dose if inside shielded area during a pulse) and neutron activation of adjacent structures (there will be very high gamma doses just after shutdown due to neutron activation).

The radiological protection aspects of a fusion facility are challenging. This is because there can be high dose rates associated with tritium, which is very light, mobile, reactive and combustible which makes maintenance more difficult. Regulations require removal of 99% of tritium in normal operations and 90% in the case of an event. There are no safety assessment principles or similar expectations for fusion.

If ITER is not ready for commissioning by 2035, a modification to the decree would need to be requested. A bilateral agreement between ASN and ITER allows ASN to access the ITER site to conduct inspections and fulfil its regulatory responsibilities.

Although ITER benefits from privileges of immunity under normal French law, it faces unique challenges due to the lack of comparable operational experience on an international scale. ITER considers lessons learned from other similar facilities while accounting for significant differences in scale. In this regard, ITER collaborates extensively with the Joint European Torus (JET) on various aspects.

There is an event reporting process for ITER and other nuclear installations in France. Duty holders are obliged by legislation to report events or incidents to ASN. ITER has to transmit on a monthly basis, any events that have happened on the construction site. Some events may be reportable on the International Nuclear Event Scale.

I–3.5.3. Preparatory study for future fusion facilities

IRSN is working on regulatory expectations/requirements for fusion. IRSN identified the following as preliminary safety issues in DEMO:

- Safety impact of the design to establish self-sufficiency for tritium;
- Long operating time and high energy neutron flux;
- Residual heat removal;
- Ionizing radiation exposure risk due to high power and neutron flux;
- Accidents to be considered, due to higher radioactive inventory and energy.

I–3.6. Regulatory inspection of fusion facilities

Article 40 of Ref. [I–28] covers inspection and enforcement. Article 29, Section III [I–28] covers the requirement to undertake a periodic review of safety every ten years.

ITER is part of ASN's wider inspection plan for regulating BNIs in France. ASN's inspections are focused on construction and the supply chain. ITER is an international organization and there is a complex supply chain. So far, ASN has undertaken inspections in the Republic of Korea, Russian Federation, India and Barcelona, making use of bilateral agreements to allow access to inspect the supply chain in foreign countries. However, there are recognized variations in safety culture across the world. These supply chain inspections are led by the local branch of ASN (Marseille).

There are approximately four one day inspections scheduled at ITER per year. This includes overseas inspections of the supply chain. If ASN is satisfied with the outcomes of these inspections, this will result in a "Lettre de suite". BNIs are not subject to licence conditions in the same way as United Kingdom nuclear licensed sites. However, the duty holder is required to demonstrate compliance with the conditions and requirements in Ref. [I–28].

ASN carries out inspections to ensure that nuclear installations are complying with the conditions set out in Ref. [I–28]. In addition, ASN sometimes undertakes themed inspections on certain issues. Topics for these themed inspections are normally identified on a risk basis, focused on higher risks and how these are controlled or based on emergent issues. ASN also undertakes inspections that focus on a particular plant or systems within an installation. This is similar to the Systems Based Inspection approach, adopted by the Office for Nuclear Regulation (ONR) in the United Kingdom. An alternative inspection approach taken by ASN is to look at the impacts of discharges and the impacts on the surroundings.

Inspections at ITER during the construction phase are focused on construction and conception related hazards and issues. Recent inspections have been checking that construction is in accordance with the specification. This has included inspections of steel rebar and concrete. Other recent inspections have been focused on the seismic pads which have recently been installed on the site or on the sectors composing the vacuum vessel.

Two of ASN's key areas of focus during the construction phase are the detritiation system, which limits releases of tritium to environment and the remote handling systems, reliable operation of which hare essential to minimise maintenance doses due to fast neutron activation of the tokamak and supporting structures.

I–3.7. Stakeholder involvement

The public is consulted before authorizing the construction of any nuclear facility. This is part of the process to establish the authorization.

I–4. GERMANY

I–4.1. Existing and planned fusion facilities

The Axially Symmetric Divertor Experiment has been operated by the Max Planck Institute for Plasma Physics since 1991, following upgrades from the original experiment setup. Its goal is to establish the physics foundation for ITER and DEMO by aligning key plasma parameters, such as density, pressure, and wall load, with the conditions expected in a future fusion power plant.

The Wendelstein 7-X, also operated by the Max Planck Institute for Plasma Physics since 2015, is the largest superconducting stellarator. It focuses on steady state plasma operation to evaluate the feasibility of the stellarator concept for a fusion power plant.

Germany has no plans to develop new fusion reactors.

I–4.2. Regulatory framework and planned regulatory approach

I–4.2.1. Regulatory framework

The German Atomic Energy Act [I–33] is not applicable as it applies solely to facilities utilizing fissile materials. The largest German fusion facility, Wendelstein 7-X, is regulated using the German Radiation Protection Act [I–34] with its subordinate Radiation Protection Ordinance [I–35]. According to these, anyone who intends to operate a plasma equipment/facility which exceeds a local dose rate of 10 µS/h, 10 cm from the walls of the area, which is inaccessible for electrotechnical reasons during operation, requires an operating licence. An accelerator or plasma facility generating more than 10^{12} neutrons per second requires an additional construction licence, because it is expected that this kind of facilities may result in radiation exposure of the surrounding population through direct or scattered radiation or through discharges of radioactive material. There are requirements to grant a construction licence, such in relation to the trustworthiness of the licence holder, the appointment of an adequate number of radiation protection officers, the limitation of the radiation exposure of the personal of the facility, the environment and the public, both during normal operation and design basis accidents.

An important difference between regulation of fission and fusion facilities is associated with the regulatory expectations for the safety cases and safety analysis that are less stringent for facilities considered as sources of ionizing radiation. The submissions are obliged to however include technical drawings, plant descriptions, details of radiation protection officers, identification of protection measures (for comparison against the state of the art), predicted radiation exposure to employees and members of the public (including contributions from direct radiation and from release of radioactive substances), security arrangements, and internal regulatory arrangements for ensuring radiation protection. The regulatory inspection and enforcement activities are commensurate with the risk presented by the facility. The licence to operate a fusion facility is issued by the competent regional authority, which usually consults with a TSO (e.g. Technical Inspection Association).

I–4.2.2. Planned regulatory approach

A decision on the approach to regulate larger fusion facilities, especially those using tritium as fuel, has not yet been made, as there are no plans to construct a large scale fusion reactor in Germany.

The TSO, GRS has published an assessment of the safety concept for fusion reactors and the extent to which existing fission regulations can be applied to fusion technologies [I–36]. GRS considers that although the risks associated with fusion are lower than for operating NPPs, it is important that a defence in depth approach (similar to the approach adopted in nuclear safety for fission reactors) is adopted to ensure that the risks associated with fusion reactors remain below an acceptable level. This study identified tritium (the most important source of hazard), dust generated and accumulated within the vacuum vessel, along with activated corrosion products, as potential sources of radioactivity during accidents. The study considers the characteristics of a number of FPPs such as ITER. It then identifies potential internal events (hazards are excluded) and calculates that:

> "the theoretically possible maximum releases from FPPs may … result in doses to the public of the order of one Sievert or below. These doses are several orders of magnitude lower than those for hypothetical worst-case scenarios of NPPs. Nevertheless, without a safety concept to exclude such releases these doses would necessitate radiation protection measures outside of the FPP".

Under contract with the European Commission, GRS, in collaboration with Karlsruhe Institute of Technology, explored how a dedicated legal and regulatory framework for fusion facilities might be structured. The project began with a review of international practices and existing regulatory approaches for fusion. This was followed by the identification of safety requirements specific to fusion, taking into account the key distinctions between fusion and fission facilities, as well as the unique safety challenges associated with fusion systems, structures, and components. Relevant documents, including European council and parliament directives, IAEA Safety Standards and other publications, and other applicable international references, were reviewed and categorized. Drawing on this analysis, the team formulated recommendations for establishing a fusion specific legal and regulatory framework and proposed an action plan for its development.

I–4.3. Technical capability in the regulatory body

In Germany, the regulatory body usually consults with a TSO on technical matters. With regard to fusion, their capability on fission safety is largely transferrable to fusion safety. To further their knowledge on fusion, GRS has, for example, established a collaboration with Karlsruhe Institute of Technology in Germany and built a network of European experts on fusion.

I–4.4. Regulations and guides including those unique to fusion facilities

There is no dedicated fusion framework. German fusion facilities are regulated using the German Radiation Protection Act [I–34] with its subordinate Radiation Protection Ordinance [I–35].

I–4.5. Licensing process

In Germany, the licensing process for facilities for the generation of ionizing radiation with a certain hazard potential (including fusion devices), typically consists of two stages: construction licence and operating licence. Both stages require submission of a safety case type document, the content of which is specified in Ref. [I–34]; however, the expectations against which these submissions will be assessed are much less formalised than for an NPP. The licensees are required to prepare a document, which includes a deterministic safety analysis. These submissions are in general less detailed than for NPPs, mostly limited to a principal level.

Germany does not have a formalised process of early engagement to regulate fusion facilities, but the regulatory body is involved from an early stage in the construction licence process. There is no formalised generic design assessment process for fusion facilities.

The licensing procedures in accordance with the German Radiation Protection Act [I–34] include the following steps:

— Licensing application;
— Examination of the application;
— Environmental impact assessment;
— Licensing Decision.

I–4.6. Regulatory inspection of fusion facilities

In Germany, the competent regulatory body of the state establishes a programme for supervisory inspections that is commensurate with the facility risk.

I–5. JAPAN

I–5.1. Existing and planned fusion facilities

The JT-60 tokamak, located at the Naka Fusion Institute, now part of the National Institutes for Quantum Science and Technology, was once among the world's largest tokamak devices. Operating from 1985 to 2008, one of its most notable contributions was supporting the design of ITER, particularly through its advancement of steady state operations for high performance plasma. JT-60 demonstrated plasma conditions equivalent to breakeven, where the input heating power and the output power from deuterium-tritium fusion reactions would be balanced, though it did not use tritium as fuel.

The JT-60 tokamak has been upgraded to the JT-60SA, a superconducting tokamak designed to further investigate steady state operation of high performance fusion plasmas, with the aim of contributing more directly to ITER and the future DEMO reactor. JT-60SA is expected to play a key supporting role in ITER research. Before ITER begins its own experiments, JT-60SA will confine high temperature plasmas at break even conditions for extended periods, providing valuable data to inform ITER operations. Unlike ITER, JT-60SA does not use tritium fuel, relying instead on hydrogen and/or deuterium. This reduces long term radioactivity, offering greater flexibility for reconfiguration as experimental experience develops. The assembly of JT-60SA for its first plasma was completed in March 2020, followed by integrated commissioning to verify the device's performance as a complete system. Improvements are being made to the electrical insulation of the joints, identified as necessary during commissioning, before the first plasma operation can proceed.

I–5.2. Regulatory framework and planned regulatory approach

Japan has no regulatory framework specifically for fusion facilities. JT-60SA, an experimental equipment relating to fusion, is regulated as a plasma generator (not as a fusion equipment) based on Ref. [I–37] and other regulations.

I–5.2.1. Legislative framework

The key legislative acts forming the foundation of the nuclear regulatory framework include:

- The NRA Establishment Act [I–38], which designates the NRA as the national nuclear regulatory body and outlines its powers and responsibilities.
- The Reactor Regulation Act [I–39], which governs all nuclear facilities and activities to protect the public and the environment. It also regulates the handling of controlled nuclear material and ensures nuclear energy is used exclusively for peaceful purposes.
- The Atomic Energy Basic Act [I–40], regarded as the cornerstone of nuclear legislation, defines "nuclear materials" as those relating to fissional materials.
- The Act of the Regulation of Radioisotopes [I–37], which sets out rules for the use, sale, rental, and disposal of radioisotopes, as well as for radiation generating devices and the management of items contaminated by radioactive substances or emissions from such devices.

I–5.2.2. Tasks and authorities of the nuclear regulation authority

The 2012 Act for Establishment of the NRA [I–38] created the NRA as Japan's regulatory body responsible for overseeing nuclear and radiation safety. Its primary objectives are to protect the public and the environment and to contribute to national security. The NRA's roles and powers are primarily set out in Refs [I–37] to [I–41].

The NRA is responsible for regulating safety, security, and safeguards in the use of nuclear energy, and safety and security in the use of radiation sources. Certain areas, such as the transport of radioactive material, medical applications of radiation, and setting of dose limits, fall under the jurisdiction of other ministries, specifically the Ministry of Land, Infrastructure and Transport and the Ministry of Health, Labour and Welfare.

In fulfilling its duties, the NRA collaborates with local authorities to conduct environmental monitoring and is authorized to issue NRA Ordinances to implement relevant laws and cabinet orders. It also holds the power to issue licences and approvals for nuclear facilities and radiation related activities, as well as to perform inspections to ensure regulatory compliance. Where non-compliance is identified, the NRA is empowered to enforce corrective actions to restore safety.

I–5.2.3. Authorization of radiation sources facilities and activities

The NRA serves as the responsible authority for authorizing the use of ionizing radiation associated with sealed sources, unsealed sources, and radiation generating devices, with certain exceptions. Specifically, the Ministry of Health, Labour and Welfare authorizes the use of radiopharmaceuticals and low energy radiation generators (below 1 MeV) in medical diagnostics. Moreover, the Ministry of Agriculture, Forestry and Fisheries handles authorizations for veterinary use, and aspects of radioactive material transport outside facilities fall under the Ministry of Land, Infrastructure and Transport.

Reference [I–37], along with its related orders and ordinances, sets out requirements for notification and authorization processes. Authorizations are granted for several categories, including Permitted Users, Specified Permitted Users, Registered Users, Permitted Waste Management Operators, Registered Users of Approved Devices with Certification Labels, and Registered Dealers or Lessors.

Applicants to the NRA provide comprehensive information, including facility plans, operational procedures for handling sources, and safety measures to prevent radiation hazards. While the NRA can attach conditions to authorizations, it does so only in exceptional cases, typically at the applicant's request, relying mainly on the detailed legislative requirements.

Once an application is approved, the applicant receives a permit for use. However, those classified as Specified Permitted Users or Permitted Waste Management Operators also undergo and pass a facility inspection, conducted either by the NRA or a Registered Inspection Body, before beginning operations. Typically, Registered Inspection Bodies carry out these inspections after NRA approval, verifying that facility arrangements comply with regulatory requirements. They issue a certificate of conformance directly to the authorized entity and subsequently provide the NRA with an inspection report.

I–6. REPUBLIC OF KOREA

I–6.1. Existing and planned fusion facilities

The KSTAR features superconducting magnets and is designed to investigate aspects of magnetic fusion energy relevant to the to the ITER project.

The Korean fusion demonstration tokamak reactor serves as a prototype for advancing commercial fusion reactor development. In its initial phase, this reactor is intended to demonstrate net electricity production and establish a self-sustaining tritium cycle, while also functioning as a facility for component testing. The second phase involves a significant upgrade, including the replacement of in-vessel components, with the aim of achieving net electricity generation of approximately 500 MWe (~2200 MW fusion power).

I–6.2. Regulatory framework and planned regulatory approach

In the Republic of Korea, the NSSC, the national regulatory body for nuclear safety, is responsible for regulating fusion facilities. Fusion facilities are subject to the regulatory framework set out under the NSA [I–42], such as the issuing of permits, initial inspection, periodic inspection, periodic reporting on the facility use. The enforcement of such regulations is undertaken by the KINS, as entrusted by the NSSC.

In the Republic of Korea, the KSTAR is operating for fusion research. In the domestic licensing process for the construction and operation of the facility, it is not regulated as a specialized fusion reactor but is managed and supervised under a regulatory framework classified as a 'Radiation Generating Device'.

Under the NSA's regulatory framework, there is no separate regulation for fusion power plants. The detailed decision for the regulatory framework to be applied to fusion power plants, such as whether they need to be regulated at the same level as fission reactors, has not yet been made. Developing the regulatory framework for fusion reactors is planned for the future.

I–6.3. Technical capability in the regulatory body

The Korean regulatory authorities have sufficient experience in regulating the KSTAR and a number of large accelerator facilities. By fully utilizing the experience, the regulatory body can consider preparing regulatory standards and procedures that reflect the unique characteristics of fusion through collaboration with the fusion communities.

I–6.4. Regulations and guides including those unique to fusion facilities

I–6.4.1. Nuclear Safety Act

The primary legislation for regulating the safety of nuclear installations is the NSA [I–42], which addresses key aspects of safety management throughout the lifecycle of nuclear energy, including research, development, production, and use.

The NSA [I–42] is the primary legislation governing the safety regulation of nuclear installations, covering key aspects of safety management across the research, development, production, and use of nuclear energy. This legal framework is organized hierarchically into five tiers:

- The NSA: The primary law enacted by the national legislature.
- The Enforcement Decree: A Presidential Decree that details the implementation of the NSA.
- The Enforcement Regulation: An Ordinance issued by the Prime Minister, providing more specific rules for applying the Act and Decree.
- NSSC Regulations: Technical standards and rules issued directly by the NSSC, such as those governing reactor facilities or radiation safety control.
- NSSC Notices: Detailed guidance and procedures published by the NSSC to supplement the regulations and standards.

The NSA includes fusion research facilities using heavy water in the classification of radiation generating devices. It is included in the same classification as the devices that generate radiation by accelerating charged particles, such as X ray generators and cyclotrons, and thus the independent regulatory system does not apply solely to fusion research facilities. However, some technical requirements are applied only to fusion research facilities, considering the unique characteristics of the facilities.

I–6.5. Licensing process

In 1995, a government led project was launched for the design and construction of KSTAR. In 2007, the main system was completed. In parallel, a regulatory study was launched to derive risk factors and to access them for the state of the art facility since 2000. The outline of the licencing process is provided below:

- In 2000: Preliminary research project for regulatory framework for KSTAR (Korea Institute of Fusion Energy and KINS);
- In 2002: Consultation on licensing issues (Ministry of Science and Technology, KFE, and KINS);
- In July 2002: Submit KSTAR licence application;
- In 2007: Seven revisions on the licence application documents;
- In December of 2007: Licence obtained.

During the application for a licence, all the documents including the radiation safety report and safety management regulations, which were required documents for the radiation generating device, not the power plant facility, were submitted to the regulatory authority of KINS.

The documentation to be submitted with an application for a licence for a radiation generator includes:

- Radiation safety report;

— Safety management regulations;
— Safety analysis report;
— Quality assurance plan.

In KSTAR’s review process, the following items were reviewed by the regulatory authority:

— The availability of legally mandated equipment and personnel;
— Satisfaction with the rules/regulations on technical standards such as radiation safety management;
— Suitability of the radiation dose (whether it satisfies proposed regulatory criteria) to workers and the public;
— Satisfaction with preparation guidelines for radiation safety report and safety management regulations.

I–6.6. Regulatory inspection of fusion facilities

The inspection of fusion research facilities is divided into facility inspection and periodic inspection. These inspections are conducted by KINS as entrusted by the NSSC. During facility inspection, whether the facility is installed in the same manner as the facility was first installed or as permitted for design changes is checked. The safety of the facility in use is checked during the periodic inspections. For KSTAR, if there is a change in what has been permitted, the facility is inspected for the change, and periodic inspections are conducted yearly.

The inspections include:

— Checking whether safety management of the facility and the workers is performed in accordance with the previously submitted safety management regulations (submission at the time of licensing and subsequent revision if necessary);
— Checking whether radiation and related risk, contamination management are being performed effectively inside and outside the facility;
— Ensuring that waste management arrangements are in place and checking records of waste management.

I–7. RUSSIAN FEDERATION

I–7.1. Existing and planned fusion facilities

In the Russian Federation, three different concepts of fusion facilities are being developed simultaneously, namely:

— Magnetic fusion (tokamaks, magnetic traps);
— Inertial fusion (laser fusion systems, accelerator driven systems);
— Hybrid fission-fusion systems.

With a timeframe of 2021–2024, the federal project Development of Controlled Nuclear Fusion and Innovative Plasma Technologies [I–43] was implemented by State Corporation Rosatom; National Research Centre, Kurchatov Institute; and the Ministry of Education and Science. The project has six main themes:

— Basic fusion technology including lithium first wall technology, operation of T-15MD tokamak;

— Hybrid fission-fusion systems including fuel pellet technology, blanket design for hybrid system;

— Innovative plasma technology including material treatment, heating systems;

— Inertial fusion technology including fusion accelerator driven neutron, ion, and other radiation sources;

— Laser fusion technology including 8 channel laser driver amplifier, laser driver;

— Regulatory frameworks including the development of regulatory framework fundamentals and regulatory documents, procedures.

The federal project also includes supporting activities which encompasses the development of information and the project exchange system.

The Russian Federation has the following fusion experimental facilities in operation:

— Tokamak-10, operated by the Kurchatov Institute since 1975;

— UNU Globus-M, operated by the Ioffe Institute since 1999 for the study of the spherical tokamak concept;

— T-15MD, a tokamak operated by the Kurchatov Institute since 2021.

I–7.2. Regulatory framework and planned regulatory approach

I–7.2.1. Regulatory framework

The term ‘fusion installation’ or similar terms are not used in the legal and regulatory framework of the Russian Federation. The definition of a ‘nuclear installation’ in the Russian Federation includes only installations that utilize fissile nuclear material. Thus, only hybrid installations that utilize a fusion neutron source for fission reaction initiation and experimental facilities that utilize nuclear material fall under this definition. Such facilities will be nuclear installations (as “other facility for the use of nuclear materials”) and are licensed in accordance with the provisions of Refs [I–44] and [I–45]. Such licences are issued by the Federal Service for Ecological, Technological and Nuclear Supervision (Rostechnadzor), which is a federal executive authority conducting safety regulation, control and supervision in the field of atomic energy use. Hybrid installations are at the design stage, and a list of obligatory safety requirements to hybrid installations is not yet established. Most of the fusion facilities in operation that utilize thermonuclear fusion reactions are granted an ionizing radiation source operating licence, since:

— Fusion installation technologies are at an early stage of development, and existing installations are used for fusion reactions and high temperature plasma confinement studies with low amounts of tritium (less than stated in federal rules and regulation in the field of nuclear energy use);

— Thermonuclear fusion studies are conducted mainly utilizing charged particle accelerators, which can provide transfer of momentum for fusion reaction realization using non-radioactive hydrogen atoms (e.g. D-D reaction).

There is a separate licensing procedure for charged particle accelerators in the Russian Federation, which are utilized not only for light hydrogen nuclei acceleration, but also heavier nuclei (e.g. for the synthesis of new chemical elements with a mass over 260 AMU).

Licensing of radiation generators is performed in accordance with the provisions of Refs [I–46] and [I–47]. Licences are issued by the Federal Service for Surveillance on Consumer Rights Protection and Human Wellbeing (Rospotrebnadzor) which is one of the federal executive authorities conducting safety regulation in the field of atomic energy use (in the area of human health).

High temperature plasma confinement processes are usually studied using non-radioactive isotopes of hydrogen and without fusion reaction realization. These facilities are not subject to licensing in the field of atomic energy use but are expected to comply with separate technical safety requirements.

However, if research of fusion reactions with tritium (for instance, deuterium-tritium reactions) or other radioactive substances is conducted at a facility, a licence is required (depending on the type of facility). Such a licence is also issued by Rostechnadzor in accordance with Refs [I–44] and [I–45].

I–7.2.2. Planned regulatory approach

There is work being carried out to develop approaches to the modification of the regulatory approach to fusion facility safety. Amongst other things, it is planned to directly include fusion installations in the list of nuclear facilities in Ref. [I–44].

The legislative framework in the field of atomic energy use does not address the hazard levels of fusion installations, however it is planned to implement a classification based on the amount of radioactive substances used and neutron radiation intensity as well as graded approach to safety requirements based on the hazard level of an installation and potential radiological consequences.

Prototype or test installations and commercial electricity generating fusion installations with the same hazard level comply with the same requirements. However, electricity generation for the general power grid would impose some additional requirements to fusion installations for commercial electricity generation including requirements regarding electricity supply to general power grid and financing of decommissioning operations.

Due to the absence of the term fusion installation, safety requirements for fusion facility regulation are not defined. In the future it is planned to establish requirements for all nuclear facilities and develop new regulatory documents to address specific aspects and hazards.

The legal framework is being developed to take into account specific features of fusion facility types, including hybrid installations. Under a federal project for the "Development of fusion technology and Innovative plasma technologies," one of the main directions includes the development of a regulatory framework for fusion. The objective is to develop regulatory support for the design, construction and operation of both fusion and hybrid fission-fusion systems, to ensure acceptable level of risks from accidents and minimize the harmful effects on workers, the public and the environment.

The work will encompass the development of the regulatory infrastructure, such as legal and regulatory amendments, and proposals for new regulations and standards. It will also include the development of scientific safety fundamentals for new fusion and hybrid system facilities, through the analysis of hazards, assessment and development of forecasting tools and codes and inspection of functioning experimental facilities.

The following timetable was established for the work:

- 2021: Identification of existing knowledge (regulatory framework, existing installations and those under construction, hazards and methods of their analysis, planning of amendments to regulatory documents and their justification);
- 2022–2023: Analytical and computational works on hazards assessments, development of codes, preparation of draft regulatory documents, experimental works and licensing support;
- 2024: Experimental, analytical, and computational works on hazards assessments, development and verification of codes, preparation of draft regulatory documents, experimental works, and licensing support.

Within the development of the regulatory framework, it is planned to conduct a comprehensive analysis of the existing Russian regulatory framework from the point of view of safety over the lifetime of a fusion facility. This will be supplemented with an analysis of international documents applicable to fusion safety, and an analysis of other nations experience in fusion regulation. Following this, several options for development of the regulatory infrastructure will be drafted, and a plan adopted. The same approach will be taken to analysing the licensing requirements for experimental fusion facilities.

The main deliverables from the project are:

- Results of implementation of fusion safety research programme (reports);
- Results of implementation regulatory infrastructure development programme (reports);
- Safety analysis for experiments at experimental fusion facilities;
- Draft regulatory documents on fusion safety;
- Draft safety cases for experimental fusion facilities;
- Verification reports for the developed fusion safety codes;
- Programme of future research on fusion safety;
- Programme of future development of regulatory infrastructure.

The general approaches to fusion safety considered are:

- The application of general nuclear safety approaches was chosen due to:
    - Radiological hazard presenting the greatest offsite threat for fusion facilities;
    - Nuclear safety methodology being arguably best described throughout various industries;
    - The same approach already practiced for ITER.

  This will require formulation of safety objectives, goals, fusion specific safety functions, application of defence in depth and identification of installation specific accident pathways.

- Presence of fusion specific hazards (electromagnetic fields, cryogenic temperatures, hydrogen as a fuel component and not a byproduct) requires thorough consideration in connection with radioactive materials present (tritium and activation products).
- Lifecycle and supporting infrastructure (e.g. tritium production and regeneration, blanket replacement modules) safety need to be taken into consideration. This is especially true for hybrid fission fusion plants taking into account the considered variety of blanket materials (e.g.

uranium, thorium), designs, and applications (energy production, fuel production, spent fuel transmutation).

— Fission based experience calls for detailed design decisions to be available prior to development of safety criteria and safety assessments, while for fusion facilities such are subject to evolution and testing. This will need to be overcome by both shifting the mentality and agreeing on the basic design features for specific (broad) types of facility.

General approaches to fusion regulations considered are:

— Regulation to be based on existing nuclear regulation infrastructure. First, amendments are required to the Nuclear Energy law to define the fusion installations.
— Decision pending on the exact classification of fusion facilities, infrastructure and relevant research facilities.
— Fusion specific safety rules, regulations and guidelines will be required, including high level (e.g. basic safety rules) and technology specific. It is too early to identify the exact number of these documents.
— The need for regulatory guidance on safety thresholds strongly expressed by the designers of fusion facilities.
— Interim regulation may be required to adapt the existing and under construction research facilities to the new safety regulation regime.
— Regulations are to be developed in close cooperation with IAEA and international fusion community (ITER project). IAEA is in good position to collect and reflect upon the available fusion safety and fusion regulation approaches taken worldwide to be used as input by national regulatory bodies.

I–7.3. Technical capability in the regulatory body

A group of experts on fusion technology and related issues work for Rostechnadzor's TSO (SEC NRS). SEC NRS works closely with scientific institutes that are the fusion technology providers and perform safety reviews of a number of fusion facilities designs. Meetings are also held with the scientific community, where proposed approach to regulation are discussed.

The regulatory body takes part in workshops and technical meetings dedicated to fusion facilities design, requirements, among other, involving design organizations and operating organizations, they are held in a proactive manner.

I–7.4. Regulations and guides including those unique to fusion facilities

The legal regulation of safety in the use of atomic energy in the Russian Federation is based on federal laws and supporting bylaws. These include:

— Legal acts enacted by the country's President and the national Government.
— Nationwide rules, standards, and regulations pertaining to atomic energy, approved by the country's primary nuclear safety regulator, complemented by materials from other agencies that monitor atomic energy utilization.

I–7.4.1. Regulations for radiation generators

With regard to radiation generators, the provisions of Ref. [I–47] establish a number of requirements for licensees, which are necessary for acquiring a licence. The licence issuing procedure is established by Ref. [I–48], and associated requirements are specified in the sanitary rules and regulations approved by Rospotrebnadzor.

I–7.4.2. Regulations for use of nuclear material and radioactive substances

For the handling of nuclear material and radioactive substances, the specific steps for obtaining a licence are outlined in the official administrative procedures governing licensing within the atomic energy sector [I–49]. Concurrently, the mandatory safety standards are detailed in two sets of regulations: federal rules specific to atomic energy use (approved by Rostechnadzor) and general public health/sanitation regulations (approved by Rospotrebnadzor). All other types of licence authorizing activities related to the use of atomic energy, including operation of radiation sources, research fission reactors, and fission reactors of NPPs, are also governed by Ref. [I–45].

I–7.4.3. Regulations related to licensing in atomic energy use

The authorization (licensing) process for activities related to the use of atomic energy is regulated by article 26 of Refs [I–44] and [I–45]. Specific requirements for the licence issuance procedure for siting, design and construction are provided in Ref. [I–49].

I–7.4.4. Regulations and regulatory guidance for siting

For siting, the following are applicable:

- Federal Law No. 170-FZ [I–44].
- Federal Law No. 174-FZ, On Ecological Expertise [I–50].
- Federal Law No. 190-FZ, Town Planning Code of the Russian Federation [I–51].
- Decree of the Government of the Russian Federation No. 280 [I–45].
- Administrative Procedures for the Public Service of Licensing of Activities in the field of Atomic Energy Use to be provided by the Federal Environmental, Industrial and Nuclear Supervision Service [I–49].

I–7.4.5. Regulations and regulatory guidance for design, construction, operation and decommissioning

The documents from Section II of Ref. [I–52] are applicable. In addition, for design, construction, operation and decommissioning, the following are applicable:

- Federal Law No. 170-FZ [I–44].
- Decree of the Government of the Russian Federation No. 280 [I–45].
- Administrative Procedures for the Public Service of Licensing of Activities in the Field of Atomic Energy Use to be Provided by the Federal Environmental, Industrial and Nuclear Supervision Service [I–49].

I–7.5. Licensing process

In accordance with Ref. [I–44], the operating organization is required to apply to Rostechnadzor for a licence to carry out specified activities during the siting, construction, operation, and decommissioning phases of nuclear facilities.

Specific requirements for the licence issuance procedure for siting, design, construction, operation and decommissioning are set out in Ref. [I–49]. Siting of installations is also regulated by Refs [I–50] and [I–51].

I–7.6. Regulatory inspection of fusion facilities

Rostechnadzor conducts audits and inspections to monitor compliance by legal entities and individuals with country's legislation, regulatory acts, and established rules and standards regarding the use of atomic energy.

I–7.7. Stakeholder involvement

Due to the fact that work on the creation of a regulatory framework for the safety of fusion facilities has just begun, there is only limited experience of interaction between Rostechnadzor TSO (SEC NRS) and future operating organizations, National Research Centre Kurchatov Institute and TRINITI JSC. Meetings and workshops are also held with the scientific community, where proposed approaches to regulation are discussed.

I–8. UNITED KINGDOM

I–8.1. Existing and planned fusion facilities

I–8.1.1. Existing facilities

JET is the only operational tokamak capable of burning tritium, which is in a 50:50 deuterium-tritium fuel mixture. Under the European Fusion Programme, JET operated from 1984 to 2023. JET does not have all the systems required of a FPP, but it does include some major fuel cycle technologies such as an Active Gas Handling System for the processing and storage of tritium. JET has three methods of heating the plasma in the toroidal chamber: (1) Ohmic Heating, (2) Neutral Beam Heating and (3) Radio Frequency Heating. JET still holds the highest peak fusion power at 16.1 MW set in 1997 and recorded the highest total amount of fusion energy produced in a plasma in 2022.

Mega Ampere Spherical Tokamak – Upgrade (MAST-U) is operated by the United Kingdom Atomic Energy Authority and started operation in 2021. MAST-U is an upgrade of an earlier Mega Ampere Spherical Tokomak, which operated from 2000 to 2013. MAST-U has several roles, one of which is to test a device known as the Super-X divertor, which is a key component in exhausting fuel not burnt in the fusion process from the spherical shaped plasma chamber and will be used for the ITER programme to confirm the design principles of its divertor.

ST40 is operated by a United Kingdom company Tokamak Energy. Tokamak Energy is planning to use the combination of spherical tokamak geometry and high temperature superconductors to achieve a very compact future FPP. Tokamak Energy is using ST40 to progress research on high temperature superconductor spherical tokomaks, one of its primary roles is to demonstrate the feasibility of the technology by achieving stable plasmas at temperatures of more than one hundred million kelvin.

First Light Fusion Ltd operates Inertial Confinement Devices where it performs experiments on inertial confinement fusion. Its approach is to fire a physical projectile at hypersonic velocity, using either a very high current electromagnetic gun or a gas gun, at a fuel pellet and from the resulting collision achieve significant levels of compression of the gas within the pellet. The compression of the pellet raises the pressure and the temperature of the gas within the pellet to the conditions necessary for fusion. Unlike many other inertial confinement devices First Light Fusion is not trying to achieve a near perfect symmetrical implosion of the fuel pellet but is instead utilizing instabilities to enhance its prospects of achieving fusion conditions.

I–8.1.2. Planned facilities and fusion power plants

Spherical Tokamak for Energy Production is a prototype FPP based on developments of spherical tokomaks. This is a publicly funded programme, with significant government support. One of its design requirements is for an electrical power output to the United Kingdom grid of ~100 MW, meaning that dependent on the efficiency of the energy conversion plant a fusion power rating of the order of up to 800 MW power may be required. The objective is to have the plant operational by 2040.

The Fusion Demonstration Plant (FDP), from General Fusion, is under design for later construction and then operation of a fusion facility that will demonstrate the company's technology for magnetized target fusion. FDP is not an FPP but a two thirds scale demonstration of the components and systems of a future FPP using General Fusion's technology. It will not use tritium in its experiments, nor will there be any power producing and conversion plant. The plan is to have FDP operational by the mid-2020s.

I–8.2. Regulatory framework

I–8.2.1. Occupational exposures framework

Workplace health and safety in the United Kingdom is regulated by three main bodies. The HSE regulates in Great Britain; the HSENI in Northern Ireland; and the ONR in Great Britain. HSE and HSENI regulate all industrial activity except for the licensed nuclear sector, which is the responsibility of ONR in Great Britain (Northern Ireland has no licensed nuclear sites). The legislative arrangements in the United Kingdom establish the licensed nuclear sector, and so ONR as the enforcing body.

All practices involving the use of radiation except for those established as licensed practices, are regulated by HSE and HSENI. Fusion facilities are not licensed and so HSE and HSENI regulate fusion in the same ways as they regulate the use of radiation for diagnostic, structural integrity, therapeutic, research and all other practices.

The nature of the radiation hazards means that other regulatory bodies ensure the risks associated are controlled and managed. The Environment Agencies in the United Kingdom regulate environmental aspects.

I–8.2.2. Regulatory framework

The Health and Safety at Work Act 1974 (HSWA) [I–53] established the principle that workplace risks are to be controlled so far as is reasonably practicable, setting out general duties for managing such risks. The HSWA adopts an outcome focused approach, requiring that risks be managed to a reasonable extent rather than prescribing absolute standards. Nevertheless, the HSWA equips the regulatory body with a variety of tools to fulfil its role in ensuring occupational health, safety, and welfare. These tools include the ability to propose new legislation in the form of regulations and to issue Approved Codes of Practice (ACOPs) and guidance.

The HSWA also enables the modernization of health and safety law within a specific legislative framework. The overarching policy is to ensure that regulations articulate general duties, principles, and objectives, while more detailed guidance is provided through ACOPs and supplementary materials. This structure is intended to minimize the need for direct regulatory intervention, promoting a proportionate and goal based approach to risk management.

The framework is structured as follows:

— General responsibilities imposed on employers, self-employed individuals, and others as outlined by the HSWA. Their scope is extensive, encompassing various individuals, locations, activities, and potential sources of hazards.
— Regulations: some elaborate upon the general obligations, while others address specific hazards or sectors. The Management of Health and Safety at Work Regulations 1999 [I–54] play a significant role by mandating employers to conduct risk assessments. These assessments help identify necessary actions to meet the HSWA's overarching requirement of controlling risks to a level that is as low as reasonably practicable.
— ACOPs offer clarifications for certain elements of the general obligations and specific regulations. Non-compliance with an ACOP can shift the burden of proof, meaning employers are to demonstrate compliance through alternative methods to avoid liability in prosecutions.
— Guidance documents, which do not hold legal authority like ACOPs but rather provide recommendations on effective measures and established best practices for risk management.

I–8.2.3. Public exposures and protection of the environment framework

Environmental regulation of industrial facilities involves separate regulatory bodies for England, Northern Ireland, Scotland, and Wales. The environment regulatory body in each of the devolved administrations regulates all industrial activities to cover both radiation facilities and nuclear licensed sites using the same regulatory framework to protect people and the environment (see Refs [I–55] and [I–56]).

On a nuclear licensed site, the environmental regulatory body regulates the disposal of radioactive waste. On a non-nuclear radioactive substances activity site, for example JET or a synchrotron such as the Diamond Light Source, the environmental regulatory body regulates the keeping and use of radioactive material and the accumulation and disposal of radioactive waste, including the physical security of the material and waste. Environmental regulation is done through the process of an operating organization obtaining either permits or authorizations from the regulatory body for its industrial activities.

I–8.3. Regulatory approach

I–8.3.1. Occupational exposure

At a high level, oversight is provided by the HSE Board—a non-executive body appointed by a United Kingdom Government Minister. The Board includes representatives from employers, employees, and other stakeholder groups such as small businesses. It is responsible for evaluating health and safety performance across Great Britain and sets out the overarching strategy to guide efforts in raising health and safety standards. The strategic plan, which spans from 2022 to 2032 [I–57] sets out five primary objectives. Among these is the goal of supporting safe innovation to prevent major incidents during the transition to Net Zero.

HSE's responsibilities include setting policy, conducting inspections, and enforcing the law when necessary. HSE regularly reviews its approach to ensure it remains effective and efficient. This overall framework gives HSE flexibility in its regulatory options and decision making depending on the hazard and associated risk. In respect of fusion technology, the United Kingdom considers the existing arrangements ensure risks to workers and others health and safety from fusion technologies are appropriately managed; however, this is subject to ongoing review.

With regard to the United Kingdom regulatory approach, first a high level assessment would consider:

- The hazards inherent to fusion and associated risk and any limitations or inadequacies with arrangements to cope with these.
- Whether putative consequences can be assigned to failures in new fusion technologies or process that would justify new arrangements.
- Public perceptions of both the hazard and the risk and whether there is an expectation there is a problem to be fixed.
- Feedback from stakeholders that arrangements are not targeted, inadequate or burdensome, confusing, or otherwise considered deficient.
- International developments and commitments mean there is an expectation HSE do something.

If this analysis shows change needs to be considered, then HSE would examine options based on the following hierarchy:

- Whether the existing domestic regulatory framework is sufficient.
- Whether in addition non-statutory guidance could be introduced to help employers develop their approach to compliance.
- Whether statutory guidance could be introduced to bring greater direction to the preferred approach to control.
- Whether new goal setting or more prescriptive regulations could be introduced specific to the hazard.

In summary, the United Kingdom approach to regulation and decision making is risk based and flexible, with greater involvement and regulatory direction as appropriate and justified by risk and inherent hazard as well as societal expectations and preferences.

I–8.3.2. Public exposures and protection of the environment approach

The regulation of nuclear and non-nuclear activities in the United Kingdom involves a comprehensive framework to manage public exposures to ionizing radiation and protect the environment. This framework encompasses environmental permitting, regulatory justification, dose limitations, siting considerations, and security measures, primarily overseen by the Environment Agency and other regulatory bodies.

- Permitting of nuclear and non-nuclear operators:

  The environmental permitting regulations [I–55] do not constitute a permitting regime comparable to that for a nuclear site licence. However, the environmental permitting regulations can contain pre-operational conditions. These conditions can be added to a permit as a regulatory hold point which would require an assessment and decision by the regulatory body before the condition can be discharged and specified operations/activities allowed to commence.

  As part of the application process, operating organizations are to show that they implement proportionate controls over their radioactive substances activities by applying the principle of optimization. The Environment Agency offers guidance to support applicants [I–58], which explains the approach to be taken, but does not prescribe specific standards. As part of the determination process, the Environment Agency assesses whether management arrangements are sufficient to achieve permit compliance, whether source security requirements are adequate and whether doses to members of the public are within specified dose constraints.

— Regulatory justification:

  Regulatory justification refers to the requirement that the overall benefits of a particular class or type of practice outweigh the potential health risks it poses [I–59]. Justification decisions are made by the government, which may consult bodies such as the Environment Agency for expert advice. The Environment Agency can only issue a permit for practices that have already been deemed justified.

  Conducting research and development by operating nuclear fission or fusion reactors is listed as a justified practice [I–60]. The Culham Centre for Fusion Energy operated by the United Kingdom Atomic Energy Authority and including the JET facility is permitted on the basis of this. The justifying authority will need to confirm whether a commercial fusion power reactor, such as the proposed Spherical Tokamak for Energy Production, is a justified practice or if a new regulatory justification decision is required. The Spherical Tokamak for Energy Production programme is developing an application to designate the operation of its fusion energy facility as a justified practice.

— Radiation protection:

  The Environment Agency is responsible for ensuring that regulated activities do not result in public exposure exceeding the prescribed dose limit. The dose limit for members of the public is 1 mSv per year. the Environment Agency also applies dose constraints to restrict the exposure of the public from individual sources. For example, a dose constraint of 0.3 mSv per year to an individual from any single source of radioactive discharge is applied. Additionally, a higher constraint of 0.5 mSv per year is applied to the total discharges originating from a single site. These limits help ensure that public exposure remains within safe and acceptable boundaries.

— Siting:

  The Environment Agency does not regulate site selection through the environmental permitting regulations [I–55]. However, it retains the right to refuse a permit application if the proposed activity would have an unacceptable environmental impact. Moreover, the Environment Agency is a statutory consultee on applications under Refs [I–61] and [I–62], and contributed to government's strategic siting assessment methodology and standards for new NPPs [I–63].

  In 2022, the government published its response [I–64] to the fusion regulation green paper, in which it proposed to develop a national fusion policy statement. This policy aims to bring the planning process for fusion energy facilities in line with that used for other nationally significant infrastructure projects and electricity generating installations. The objective is to create a more streamlined and efficient planning framework for future fusion developments.

— Security:

  The Environment Agency works alongside counter terrorism security advisers within the local police forces to ensure suitable security arrangements are in place. These requirements do not apply on nuclear licensed sites where ONR is the security regulator.

Experimental fusion facilities such as JET are not classified as Nuclear Installations and consequently, they neither require a nuclear site licence nor are regulated for safety by ONR. The United Kingdom's Government will use the Energy Security Bill [I–65] to introduce legislation clarifying that fusion energy facilities will not be classified as nuclear installations under the law. There is no legal definition of a fusion facility or fusion power plant in the United Kingdom. For example, JET has a Consent to operate from HSE as an accelerator.

I–8.3.3. Future public exposures and protection of the environment approach

In 2021, the United Kingdom government published a suggested approach for a regulatory framework for fusion energy [I–66]. This proposed to regulate FPPs as radiation facilities, not as nuclear installations. This means that HSE will retain its regulatory responsibility for the regulation of all fusion facilities including all types of fusion energy system such as FPPs.

The government has also determined that the regulatory framework is suitable for overseeing fusion energy facilities, with the Environment Agency and HSE continuing their roles. Following a consultation, the government outlined plans to address key issues, i.e.:

- The designation of fusion energy as a justified practice;
- A national fusion policy statement will be developed to streamline planning processes;
- The government will explore establishing a liability regime for fusion operators to cover third party claims from accidents;
- The Energy Security Bill will clarify that fusion facilities are not legally classified as nuclear installations to avoid inconsistencies;
- Regulatory bodies will work with industry and international experts to create fusion specific engagement processes and guidance to clarify regulatory obligations for developers.

I–8.3.4. Environment Agency collaboration with other United Kingdom regulatory bodies

The Environment Agency consults organizations or bodies that possess relevant expertise or local insight, such as local authorities and other government agencies and public bodies. The Environment Agency has established formal working relationships with specific organizations through joint agreements such as working together agreements and memorandums of understanding. Relevant memorandums of understanding include agreements with the ONR concerning work on nuclear licensed sites and the transport of radioactive material [I–67], and with the HSE for activities involving other radioactive substances [I–68].

I–8.3.5. Safeguards

Fusion facilities in operation in the United Kingdom do not require safeguards regulation as they do not use or store nuclear material. However, as the Culham Centre for Fusion Energy, where JET is located, does utilize, for example, tritium from Canada, this site is subject to safeguards inspection by ONR. ONR also undertakes safeguards inspections at non-nuclear radioactive substances activities sites, and transport regulations inspections.

I–8.4. Regulatory inspection of fusion facilities

I–8.4.1. Occupational exposure

The inspection of fusion facilities covers all aspects of occupational health and safety and is carried out in support of ensuring compliance with HSWA and the associated regulations. Inspectors have a right of entry and other powers available to them through for example enforcement of poor compliance through use of improvement and prohibition notices. Inspection is focused on the radiation hazard, especially in relation to compliance with Refs [I–69] and [I–70]. Regulations require a graded approach to radiation risk: fusion facilities are required to gain consent for the operation of an accelerator and, if quantities of radioactive discharges exceed thresholds, discharge of radioactive substances.

Existing fusion facilities in the United Kingdom are regarded as presenting no greater risk to health and safety than other large industrial facilities. However, fusion power plants are likely to present a different health and safety challenge, which may mean different inspection programmes and approaches are necessary.

I–8.4.2. Public exposures and protection of the environment

The Environment Agency conducts independent inspections (planned and unplanned, with or without prior notice) to ensure operating organizations meet all legal requirements, including the specific conditions set out in their permits in relation to protection of the public and the environment. Inspection activities can take the form of site inspections, audits, check monitoring/sampling as well as reviewing reports, data, procedures and management system of the operating organizations. When developing its inspection plans, the Environment Agency adopts a graded approach, considering the potential scale and characteristics of hazards at the site.

I–8.5. Employer and public engagement

I–8.5.1. Occupational exposure

HSE determines good practice through a collaborative process, reaching consensus by engaging in discussions with stakeholders, including employers, other government bodies, and health and safety experts. International developments, as appropriate, are also relevant. The aim is to agree what is reasonable and proportionate rather than technically possible to avoid placing undue burdens on employers.

The HSWA mandates formal consultation on certain proposals, new regulations, and statutory guidance. More generally, as a regulatory body HSE signs up the United Kingdom government's principles for consultations with the public and others.

I–8.5.2. Public exposures and protection of the environment

The Environment Agency records on a publicly available register all applications for new permits, variations, transfers, or surrenders. Exceptions are made for national security and commercial confidentiality. For sites of high public interest, the Environment Agency may, by agreement and consistent with its communications strategy, make the application and draft decision document available at public places, such as council offices or libraries. The Environment Agency is required to consult on all applications for new permits.

If the Environment Agency decides to undertake additional consultation, a consultation strategy is prepared in advance. The extent of this additional consultation depends on the significance of the site and the nature of the facility being permitted. While the major element of consultations will be on the basis of the written documents, the Environment Agency may decide to supplement these with public meetings, public surgeries, or drop-in sessions involving the applicant and interested parties. When deciding on consultees, the Environment Agency consider those who have responded on previous occasions or have expressed interest in any pre-application publicity, since they may legitimately expect to be consulted.

It is almost certain that any new FPP will be a part of the United Kingdom's nationally significant infrastructure projects. This will include formal provisions for a wide ranging consultation process informed by a public engagement process.

I–9. UNITED STATES OF AMERICA

I–9.1. Existing and planned fusion facilities

There are several projects in the United States of America focused on fusion energy. First, the Tokamak Fusion Test Reactor, was designed to reach breakeven energy output compared with energy input (although this was not achieved). It used tritium and deuterium fuel, and it was the first to generate over 10 MW of fusion power. Second, there is DIII-D research programme that aims to ascertain the scientific principles requisite for the optimization of the tokamak methodology in the context of fusion energy generation. In addition, the National Spherical Torus Experiment was designed to study the physics of spherically shaped plasmas. Lastly, the National Ignition Facility, found at the Lawrence Livermore National Laboratory in Livermore, California, functions as a large scale, laser powered, research apparatus for inertial confinement fusion. It utilizes lasers to warm and compress a small volume of hydrogen fuel, with the intention of initiating fusion reactions. The National Ignition Facility's primary goal is to attain fusion ignition alongside a substantial energy gain.

There are at least twelve private companies headquartered in the United States of America seeking to develop and deploy fusion energy technologies. These include Commonwealth Fusion Systems, CTFusion, Helicity Space Corporation, Helion Energy, Horne Technologies, HyperJet Fusion Corporation, LPPFusion, Magneto-Inertial Fusion Technologies, Princeton Fusion Systems, TAE Technologies, Type One Energy Group, and Zap Energy. Several companies have prototype/pilot facilities under construction or planned. Commercial fusion power plants are scheduled for deployment in the 2030s.

I–9.2. Regulatory framework

The US NRC's regulatory responsibilities encompass the licensing and oversight of nuclear reactors, fuel cycle facilities, and other civilian applications of radioactive material. This includes nuclear medicine programmes in hospitals and academic pursuits at educational and research establishments. Furthermore, the US NRC applies these same functions to regulate industrial uses, such as gauges, irradiators, and other apparatus containing radioactive material. The US NRC's regulations and practices consider the magnitude of hazards and other contributors to the risks posed by radioactive material in determining the appropriate controls and oversight of the licensed activities. Considering this framework, the US NRC is working on devising appropriate regulatory strategies for fusion facilities (see Refs [I–71] to [I–76]). In 2009, the US NRC recommended the establishment of its jurisdiction over fusion facilities. This was agreed, but the question on how to regulate fusion facilities was deferred until the technology developed further.

Research facilities related to the development of fusion technologies are not licensed or regulated by the US NRC beyond their possible possessing of radioactive material, such as tritium. Government sponsored fusion research may fall under the jurisdiction of safety orders issued by the DOE. Other federal, state, or local laws and regulations established to protect workers and the environment from various hazards may also be applicable to fusion research.

I–9.3. Regulations and guidance

While the United States of America has no specific regulatory framework for fusion facilities, its nuclear and radiation safety laws and regulations are structured hierarchically. This three level system includes foundational nuclear safety acts, detailed federal regulations for implementation, and complementary regulatory guidance. Furthermore, industry consensus standards for nuclear and radiation safety can be incorporated into this regulatory structure by formal endorsement in federal regulations or guidance.

The American Society of Mechanical Engineers Boiler and Pressure Vessel Committee, specifically its Section III on Construction of Nuclear Facility Components, is actively working to establish guidelines for building fusion energy devices. The Standards Committee of Section III, Division 4, and its Subgroup on Fusion Energy Devices are responsible for creating these new fusion Code rules. A draft version for trial [I–77] was issued in 2018 and is under review, with significant future amendments expected. These rules cover fusion energy related parts such as vacuum vessels, cryostats, and superconducting magnet structures, along with their interdependencies. Furthermore, they extend to associated support structures (both metallic and non-metallic), containment structures, and internal vessel components such as fusion system pipework, vessels, valves, pumps, and supports. The guidelines set out comprehensive requirements for materials, from design and manufacturing to stamping.

I–9.4. Regulatory inspection

I–9.4.1. Regulatory inspection of nuclear power plants

Under the Atomic Energy Act, the US NRC is authorized to inspect NPPs to protect the public, as well as national defence and security. US NRC staff conduct inspections of NPPs during construction, testing, and operation to ensure compliance with regulations and licence conditions. This inspection programme assesses whether activities are being carried out correctly and equipment is properly maintained, verifying that licensees are operating their facilities safely. The inspection findings are integrated into the agency's overall evaluation of a licensee's performance. The US NRC can also take enforcement action to address any safety and security concerns or breaches of its requirements.

All inspection findings are formally documented, with inspection reports typically issued quarterly for each NPP. Furthermore, senior US NRC officials review the performance of plants with identified issues during the annual agency action review meeting. The outcomes of this review are presented at a public meeting, offering a forum to discuss significant incidents, performance trends, and measures to prevent future occurrences.

I–9.4.2. Regulatory inspection of fusion facilities

The US NRC is developing the regulatory approach for commercial fusion energy systems. Matters related to licensing, regulation, inspection, and enforcement will be addressed within the activities and related regulatory infrastructure development, which is expected to be completed by 2027.

I–10. EUROPEAN UNION

I–10.1. European Union and Euratom law and legislation

In the European Union, legislative acts are initiated by the European Commission and are adopted either by both the Council of the European Union and the European Parliament or, in some cases, solely by the Council, following procedures outlined in the European Union treaties.

Among the foundational treaties of the European Union is the Treaty establishing the European Atomic Energy Community (Euratom) [I–78], which remains in force and applies exclusively to the civilian use of nuclear energy. The Euratom Treaty aims to support a common nuclear energy market with objectives that include promoting research, ensuring a secure supply of nuclear materials across European Union Member States, and establishing a robust oversight system to guarantee the peaceful use of nuclear materials and uphold high health and safety standards.

Although the Treaty does not explicitly define 'nuclear material', it categorizes it as "ores, source materials, and special fissile materials," with detailed definitions provided in Article 197 of the Consolidated Version of the Euratom Treaty [I–78].

The European Union promotes the above objectives through:

- The European Union's radiation protection legislation encompasses several key directives and regulations to ensure safety from ionising radiation and radioactive substances. These include the Basic Safety Standards Directive [I–79], which establishes uniform standards for protecting workers, the public, and patients from ionising radiation; the Drinking Water Directive [I–80], which sets requirements for monitoring radioactive substances in water intended for human consumption; and the Directive on the Supervision and Control of Shipments of Radioactive Waste and Spent Fuel [I–81]. Additionally, the Council Regulation on Shipments of Radioactive Substances between Member States [I–82] governs intra-European Union transfers, while the Council Regulation on Maximum Permitted Levels of Radioactive Contamination in Food and Feed Following A Nuclear Accident [I–83] sets safety thresholds. Decisions on early notification and assistance in nuclear or radiological emergencies are covered under Refs [I–84] and [I–85], respectively. Further information is available on the European Commission's radiation protection website [I–86].
- European Union legislation on the management of radioactive waste and spent fuel, including the Radioactive Waste and Spent Fuel Management Directive [I–87]. More details are available on the European Commission website for radioactive waste and spent fuel [I–88].
- European Union legislation on nuclear safety, including the establishing a community framework for the nuclear safety of nuclear installations [I–89]. More details are available on the European Commission website for nuclear safety [I–90].
- European Union legislation on the decommissioning of nuclear installations (i.e. power plant or research reactor), including to support decommissioning programmes for 2021–2027. More details are available on the European Commission website for decommissioning of nuclear facilities [I–91].
- European Union legislation on nuclear safeguards to ensure that nuclear material is used only for peaceful purposes, including the Regulation on the application of Euratom safeguards. More details are available on the European Commission website for safeguards [I–92].

I–10.2. Interactions with European Union national regulatory bodies

The European Commission engages with European regulatory bodies through the European Nuclear Safety Regulators Group [I–93]. This independent group of experts consists of senior figures from national nuclear safety, radioactive waste safety, or radiation protection regulatory authorities, along with senior civil servants from all European Union Member States who have expertise in these areas, and European Commission representatives.

I–10.3. European Commission activities concerning the regulatory framework for fusion energy

In 2020–2021, a review was undertaken to explore the available regulatory options for fusion power plants [I–94] and in 2022, the Euratom Research and Training Programme launched the HARMONISE grant [I–95] towards the harmonization in licensing of future innovative fission and fusion power technologies in Europe. In 2022, EUROfusion[6] set up a working group involving several nuclear safety experts in the fusion and fission fields to draw recommendations on the first safety principles for the development of a regulation tailored to fusion power plants.

The Study on the Applicability of the Regulatory Framework for Nuclear Facilities to Fusion Facilities [I–97] identified fusion specific safety issues and proposed safety concepts for fusion facilities and an action plan to establish a fusion specific regulatory framework. It was concluded that European secondary legislation does not address specific requirements for fusion facilities. However, the following would be applicable to fusion power plants:

— Council Directive 2013/59/Euratom – Basic Safety Standards [I–79];

— Directive 2011/70/Euratom – Radioactive Waste and Spent Fuel Management [I–87];

Other secondary legislation applies only for nuclear facilities, meaning that fusion power plants are out of scope.

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ANNEX II.
KEY CONSIDERATIONS FOR EFFECTIVE REGULATION OF FUSION FACILITIES

II–1. WASTE MANAGEMENT

Waste produced by fusion facilities includes radioactive and non-radioactive waste. There are no challenges considered for non-radioactive waste beyond those already encountered in other industries. Additional information is needed by regulatory bodies on the types and quantities of radioactive waste to be generated by fusion facilities to evaluate whether updated or new regulatory measures and/or guidance are needed.

Radioactive waste at fusion facilities will include structures and components that become radioactive or become contaminated with tritium. The quantities and activity of the waste need to be known by regulatory bodies and will be dependent on design and technology choices made by the developers, for example, the inclusion of impurities with special activation potential in structures and components. The development of novel materials for fusion facilities will need to take into account waste management options to avoid waste with no disposal route or waste that will require extended storage periods (e.g. activated impurities could result in longer storage periods). The waste management options will also need to consider waste treatment technologies that might not already exist.

Future fusion facilities will produce a large volume of low level waste, which may be considered for recycling. The opportunity to treat and recycle the low level waste is being explored and would need to be accepted by a downstream user (such as the fission industry) as the material will likely remain active.

II–2. TRITIUM INVENTORY AND MANAGEMENT

The projected on-site tritium inventory for a fusion power plant is in the order of a few kilograms, equivalent to approximately 4×10^{18} Bq. This amount is considerably larger than the roughly 100 g (4×10^{16} Bq) of tritium used at the JET facility. There are issues associated with handling large amounts of fuel and as such the safe handling and storage of tritium will be required. Tritium is also highly mobile in the environment so the exposures from misuse or an accident could be significant. The exposure from activated corrosion products, tritiated dust, and radionuclides also needs to be considered and could be significant.

Existing regulatory requirements need to be evaluated so that appropriate confinement measures are in place to ensure that tritium releases remain within regulatory limits and to ensure that appropriate emergency plans are in place.

II–3. ENGINEERING CHALLENGES

There are multiple engineering challenges associated with fusion facilities that need to be understood by regulatory bodies. The environment in a fusion facility can include very high temperatures, high heat flux, and exposure to high energy neutrons, which are beyond those experienced in the fission industry.

High energy neutrons up to 14 MeV cause atomic displacement in the surrounding wall of the reactor. Also, high energy neutrons interact with the nucleus of the atom and cause transmutation of the material which can cause different impacts. Interactions are severe and can radically change the properties of the material and can cause embrittlement and material swelling from gaseous transmutation. Engagement with fusion researchers and designers will help to understand the challenges presented by high energy neutrons on structures, systems, and components and how the materials can be qualified.

Other aggressive environmental factors include magnetically imposed stresses and high mechanical load, and plasma effects causing erosion or ablation. Designers are planning to use exotic materials for coolant, breeders and multipliers. How such materials respond to a fusion environment is unknown. The fusion industry does not have the modelling capability to fully understand the material response.

Some fusion technologies, including a variety of inertial fusion concepts, use a very high velocity projectile fired by a gas gun to initiate fusion. The projectile collides with a fusion fuel bearing target. The target serves to focus the projectile's energy, which then implodes the fuel, compressing and heating it to the conditions needed for fusion. Regulatory bodies need an awareness of the safety of the very high velocity projectile and the large amounts of propellant to fire the gas gun.

Research fusion facilities typically produce limited residual heat upon plasma shutdown. For instance, ITER is expected to generate 11 MW at shutdown, which reduces to 0.6 MW after a day. However, demonstrator facilities and FPPs are projected to have significantly greater residual heat due to their extended operating times. The European DEMO, for example, is anticipated to produce 80 MW at plasma shutdown, decreasing to 30 MW after a day. Similarly, during maintenance for the former Japanese SlimCS DEMO reactor project, one sector was estimated to have 4.55 MW of residual heat at shutdown, falling to 0.26 MW after one month. This particular sector required cooling during transport, as without it, its temperature would reach approximately 1000°C after about 40 days.

Plans for the European DEMO include provisions for storing internal vacuum vessel components for a cooling phase estimated to last about 18 months. The consequence of failure of cooling during transportation within the facility needs to be examined.

II–4. LACK OF OPERATING EXPERIENCE AND RELEVANT GOOD PRACTICE

A number of fusion designs are at a low technology readiness level[7] and are in prototype and concept design stages. As such, there is limited relevant operational experience available for fusion developers to reference during the safety demonstration of their designs. The use of modelling and digital twins could be used to demonstrate the safety of fusion research facilities and, as the designs develop, the models and digital twins can be validated by operational experience at an appropriate time. The models and digital twins for fusion facilities will need to be validated and regulatory bodies will need the associated knowledge to assess the models and digital twins to understand the conservatism in the simulations.

II–5. PUBLIC ENGAGEMENT ON APPLICATIONS

Public engagement during the development of a national regulatory framework may provide regulatory bodies with additional insights on the expected effectiveness of the proposed framework and identify unintended consequences or regulatory gaps prior to implementation. Such engagement

[7] For more information on the application of technology readiness levels to fusion technology, see IAEA-TECDOC-2047, Considerations of Technology Readiness Levels for Fusion Technology Components (2024): https://www.iaea.org/publications/15615/considerations-of-technology-readiness-levels-for-fusion-technology-components.

may also benefit developers and industry groups by promoting early understanding of planned regulatory requirements, which may influence the design of future fusion facilities.

Engagement with the public on the authorization of fusion facilities by regulatory bodies and developers would be beneficial for public acceptance. Public engagement is conducted in various forms by regulatory bodies and developers, from sharing design and regulatory assessment documents to conducting face-to-face question and answer sessions. The timing and scale of the public engagement during the authorization process would need to be considered to maximise the benefit to the public, developers, and regulatory bodies.

II–6. REGULATORY EXPERIENCE OF DESIGNERS

There are numerous fusion designers with a wide range of experience. Some designers have limited experience and would benefit from early engagement with regulatory bodies to explain the regulatory framework. Designers with limited experience may benefit from working with experienced TSOs who can provide an advisory role in the development of applications. Other designers may have a more sophisticated understanding of regulatory requirements from previous engagements with regulatory bodies on the authorization of nuclear facilities. However, further engagement with the regulatory body will be necessary to understand new or different requirements established for fusion facilities.

LIST OF ABBREVIATIONS

ACOP	Approved Code of Practice
AMT	advanced manufacturing technology
ASIPP	Institute of Plasma Physics of Chinese Academy of Sciences
ASN	Nuclear Safety Authority (Autorité de Sûreté Nucléaire)
BNI	basic nuclear installation
CFETR	China Fusion Engineering Test Reactor
CNPE	China Nuclear Power Engineering Company
CNSC	Canadian Nuclear Safety Commission
DEMO	DEMOnstration Power Plant
DOE	United States Department of Energy
Euratom	European Atomic Energy Community
FPP	fusion power plant
GF	General Fusion
GRS	Gesellschaft für Anlagen- und Reaktorsicherheit
HSE	Health and Safety Executive
HSENI	Health and Safety Executive for Northern Ireland
HSWA	Health and Safety at Work Act
IRSN	Institut de radioprotection et de sûreté nucléaire
JET	Joint European Torus
KINS	Korea Institute of Nuclear Safety
KSTAR	Korea Superconducting Tokamak Advanced Research
MAST-U	Mega Ampere Spherical Tokamak - Upgrade
NEIMA	Nuclear Energy Innovation and Modernization Act
NNSA	National Nuclear Safety Administration
NPP	nuclear power plant
NRA	Nuclear Regulation Authority
NSC	Nuclear and Radiation Safety Center, China
NSA	Nuclear Safety Act
NSCA	Nuclear Safety and Control Act
NSSC	Nuclear Safety and Security Commission
ONR	Office for Nuclear Regulation
SEC NRS	Scientific and Engineering Centre for Nuclear and Radiation Safety
SWIP	Southwestern Institute of Physics
TSN	Transparency and Nuclear Safety Act
TSO	technical support organization
VDR	Vendor Design Review
US NRC	United States Nuclear Regulatory Commission

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